

How Nonlinear Is It? A Tutorial on Nonlinearity of Orbit and Attitude Dynamics

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Abstract

Some observations are made on coordinate selection for the two most fundamental problems of astrodynamics, namely orbit dynamics and attitude dynamics, and some interesting connections and analogies between the two problems are explored. While an infinity of coordinate choices are feasible for each of these problems, we review four coordinate choices for each problem, including several that lead to governing differential equations that are regular, and in some cases, rigorously linear. Some methodology is introduced that has a universal flavor with implications for dynamical systems broadly. We show how dramatic qualitative and quantitative alteration of the mathematical description of the motion can be accomplished by introduction of redundant coordinates to describe the evolution of the dynamics in a higher dimensioned space. Some observations are made on an analogy between the two central problems of interest. One of the regularizing transformations studied for orbital dynamics is motivated directly by this analogy. Finally broadly applicable analytical and computational developments are presented that provide a measure of nonlinearity over a worst-case region of state space in the vicinity of a reference trajectory; this measure is used to evaluate the several coordinate choices.

Introduction

As a gross over-simplification, much of the history of analytical mechanics, especially celestial mechanics, can be characterized as a quest to find judicious coordinates in particular problems, or to establish broadly applicable methodology for classes of problems. This quest led Hamilton and Jacobi to develop canonical transformation theory [1], Euler and Hamilton to discover the quaternion, Lagrange and Poisson to invent the method of variation of parameters [2], Cayley and Rodrigues to introduce novel representations of rotations [3], and many modern analysts to formulate regularizing coordinate transformations with important consequences in dynamical systems broadly and astrodynamics in particular. In the classical works,

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usually one sees the motivation for coordinate transformation theory as mainly to establish canonical coordinates wherein exact integrals of the system can be easily obtained, or wherein separability (in the sense of Hamilton-Jacobi theory [1]) can be achieved, or wherein approximations such as averaging are easily carried out.

A favorite parlor game of us engineers, scientists and mathematicians is discussing “how nonlinear” is our system du jour. In many cases, such qualitative discussions fail to focus on an important underlying truth: nonlinearity is not an inherent attribute of a physical system, but rather is heavily dependent upon our *mathematical description* of the system’s geometry, kinematics, and evolution dynamics. With fixed physical model assumptions, an infinity of coordinate choices is typically possible, and it is apparent that the issue of analyst decisions (which are frequently made subjectively!) cannot be dismissed as immaterial to the system nonlinearity. Broadly speaking, it should be easily recognized that the nonlinearity of any nontrivial dynamical system is strongly coupled to the decisions made when establishing a mathematical description of the physical system, and *not an attribute intrinsic to the system itself*. While it is true that most nonlinear systems cannot be linearized simply by judicious transformations, it is vitally important that a significant number of the problems in astrodynamics can in fact be rigorously linearized (*without approximation*) and regularization (elimination of singularities) can even more broadly be accomplished merely by choosing judicious coordinates. There is no universal consensus on what we mean when we say “System A is *more nonlinear* than System B,” so a portion of this paper will be devoted to this issue: How do we measure nonlinearity?

A Broadly Applicable Approach to Measuring Nonlinearity

Nonlinearity Measure for Algebraic Systems

First we consider the nonlinear algebraic system

$$\mathbf{y} = \mathbf{f}(\mathbf{x}), \mathbf{y} \in \mathcal{R}^m, \mathbf{x} \in \mathcal{R}^n, \mathbf{f}(\mathbf{x}) \text{ is a differentiable vector-valued function} \quad (1)$$

Suppose that the map $\mathbf{y} = \mathbf{f}(\mathbf{x})$ is considered locally in the vicinity of the expected $\mathbf{x}$, $\bar{\mathbf{x}} \triangleq E(\mathbf{x})$. Specifically consider the family of finite $\mathbf{x}$ -variations: $\mathbf{x} = \bar{\mathbf{x}} + \delta\mathbf{x}$ and how they map through equation (1) into the corresponding finite $\mathbf{y}$ -variations:

$$\mathbf{y} = \mathbf{f}(\bar{\mathbf{x}} + \delta\mathbf{x}) = \bar{\mathbf{y}} + \delta\mathbf{y}, \bar{\mathbf{y}} \triangleq \mathbf{f}(\bar{\mathbf{x}}) \quad (2)$$

Frequently, we can bound the set of worst-case $\delta\mathbf{x}$ variations as

$$\|\delta\mathbf{x}\| \leq \delta x_{\max} \quad (3)$$

Over the worst-case ball of equation (3), of radius $\delta x_{\max}$ and centered on $\bar{\mathbf{x}} \triangleq E(\mathbf{x})$, the globally nonlinear function of equation (1) may or may not be locally near-linear over this region, and in any event, we are interested in measuring the validity of the local linear approximation

$$\delta\mathbf{y} = \mathbf{f}(\bar{\mathbf{x}} + \delta\mathbf{x}) - \mathbf{f}(\bar{\mathbf{x}}) \cong \bar{A}\delta\mathbf{x}, A(\mathbf{x}) = \left[\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} \right], \bar{A} = A(\bar{\mathbf{x}}) = \left[\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} \right]_{\bar{\mathbf{x}}} \quad (4)$$

More generally, we may have some other nonspherical definition for the set of worst-case $\delta\mathbf{x}$ -variations, such as the set of points lying on the surface of a $k - \sigma$ ellipsoid [4] associated with the uncertainty of the $\delta\mathbf{x}$ -variations. In general, we can use an arbitrary process to define the set of worse cases $\{\delta\mathbf{x}_1, \delta\mathbf{x}_2, \delta\mathbf{x}_3, \dots, \delta\mathbf{x}_N\}$,

however, for this tutorial, we are mainly interested in these variations to being near-uniformly spaced on a worst-case surface, e.g., an n -dimensional sphere with $\|\delta \mathbf{x}_i\| = \delta x_{\max}$. More generally, the worst-case surface could be an ellipsoid, but an ellipsoid can be mapped into a sphere by judicious coordinate rotation and scaling. The departure of the Jacobian matrix A from being constant over the family of worst-case variations is an intuitively attractive way to measure nonlinearity. In reference [4], we introduced the nonlinearity index

$$\nu \triangleq \sup_{i=1, \dots, N} \frac{\|A(\mathbf{x}_i) - A(\bar{\mathbf{x}})\|}{\|A(\bar{\mathbf{x}})\|} \quad (5)$$

The norm used in equation (5) is the Frobenius norm (sum square of the elements). Note that matrix norms are most meaningful if $\mathbf{y}$ and $\mathbf{x}$ are nondimensionalized so that all elements of both $\mathbf{y}$ and $\mathbf{x}$ are of order one. Then equation (5) is itself a nondimensional measure of the variation of the Jacobian over a set of extreme points; it is a *matrix relative error* of the largest variation found in the Jacobian. Note that a ν of order 10^{-2} or smaller qualitatively indicates “less than 1%” nonlinearity of equation (2) when mapping worst-case variations in the vicinity of $\bar{\mathbf{x}}$, over the finite sample on the boundary of the region of equation (3). Further note that a ν of order one indicates qualitatively a “100%” worst-case departure from the linear approximation. The utility of this index is of course not deterministic, but, for example, in [4], it is shown to be an excellent indicator of when we can approximate error propagation statistics using linear covariance analysis. When generalized (below) for dynamical systems, this idea is also useful when we wish to select the most nearly linear set of coordinates for a given dynamical system, considering the finite region associated with worst-case variations. The case of a scalar function of two variables is shown in Fig. 1. Note the fractional variation of the gradient over a family of worst variations is an intuitively reasonable measure of nonlinearity.

We further note that this measure has been found to work best, for finite N , if the worst-case points are equally spaced over the domain of the worst-case variations.

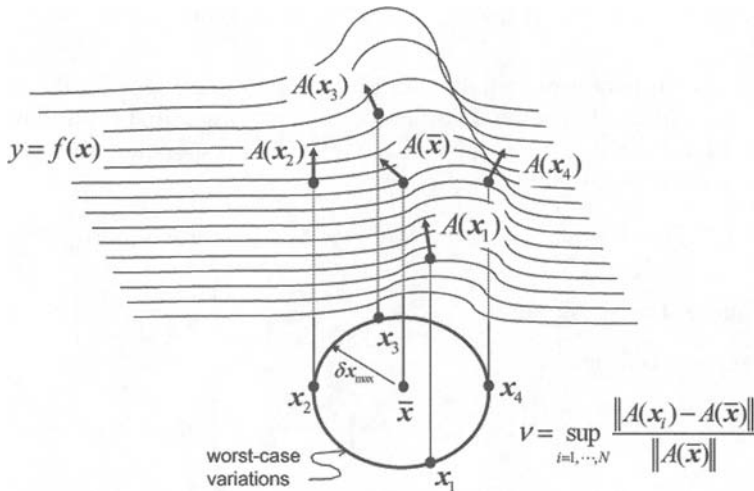


FIG. 1. Worst-Case Variations: A Nonlinear Scalar Function of Two Variables.

Recent research has led to a general purpose approach to constructing N approximately uniformly spaced points on an n -dimensional surface, see references [3, 5–7]. It should be noted that the problem of uniformly placing points on a sphere is generally not exactly solvable. For the case of $n = 3$, it is well known [7, 8] that only for $N = 4, 6, 8, 12$, and 20 (the five regular polyhedra) it is possible to exactly place uniformly spaced points on a sphere. However, it has been found that, for larger N , the problem can be solved to a high degree of approximation judged to be more than sufficient for the present discussion. Furthermore, the measure of equation (5) has been found to converge much more quickly than does for Monte Carlo methods. This seems logical, since all samples near-uniformly cover the worst-case surface, instead of statistically populating the whole space. In any event, the number N of worst-case variation points should be increased until ν converges acceptably. We will find that the above developments must be extended and modified for the case that the system is subject to constraints; we will make use of differential constraint elimination so that the Jacobian is informed of the constraint. Furthermore, additional generalizations are required to accommodate differential equation models.

Nonlinearity Measure for Dynamical Systems

The above nonlinear algebraic ideas readily generalize to systems described by nonlinear differential equations. Consider the initial value problem

$$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}), \mathbf{x}(t_0) = \mathbf{x}_0 \quad (6)$$

In equation (6), $\mathbf{f}(t, \mathbf{x})$ is a differentiable vector-valued function and $(\cdot) \triangleq d(\cdot)/dt$. Let $\bar{\mathbf{x}}(t_0) = E\{\mathbf{x}_0\}$ be the expected initial state and $\bar{\mathbf{x}}(t)$ be the trajectory that ensues from solving equation (6) with the expected initial conditions. The corresponding state transition matrix evaluated along $\bar{\mathbf{x}}(t)$ is denoted

$$\bar{\Phi}(t, t_0) = \left[\frac{\partial \bar{\mathbf{x}}(t)}{\partial \bar{\mathbf{x}}(t_0)} \right] \triangleq \left[\frac{\partial \mathbf{x}(t)}{\partial \mathbf{x}(t_0)} \right]_{\bar{\mathbf{x}}(t)} \quad (7)$$

which satisfies

$$\dot{\bar{\Phi}}(t, t_0) = \bar{A} \bar{\Phi}(t, t_0), \bar{A} = \left[\frac{\partial \mathbf{f}(t, \bar{\mathbf{x}}(t))}{\partial \bar{\mathbf{x}}(t)} \right], \bar{\Phi}(t_0, t_0) = I \quad (8)$$

Consider a distribution of initial conditions neighboring the expected initial state $\bar{\mathbf{x}}(t_0) = E\{\mathbf{x}(t_0)\}$. Each member of a family of worst-case initial condition variations $\mathbf{x}_i(t_0) = \bar{\mathbf{x}}(t_0) + \delta \mathbf{x}_i(t_0)$, $\|\delta \mathbf{x}_i(t_0)\| = \delta x_{\max}$ yields a corresponding trajectory denoted $\mathbf{x}_i(t)$, satisfying the differential equation

$$\dot{\mathbf{x}} = \mathbf{f}(t, \mathbf{x}), \mathbf{x}(t_0) = \mathbf{x}_i(t_0) = \bar{\mathbf{x}}(t_0) + \delta \mathbf{x}_i(t_0), \text{ with } \|\delta \mathbf{x}_i(t_0)\| = \delta x_{\max} \quad (9)$$

The initial state sensitivity $\Phi_i(t, t_0) = \left[\frac{\partial \mathbf{x}_i(t)}{\partial \mathbf{x}_i(t_0)} \right] \triangleq \left[\frac{\partial \mathbf{x}(t)}{\partial \mathbf{x}(t_0)} \right]_{\mathbf{x}_i(t)}$ of each trajectory is generated from

$$\dot{\Phi}_i(t, t_0) = A_i \Phi_i(t, t_0), A_i = \left[\frac{\partial \mathbf{f}(t, \mathbf{x}_i(t))}{\partial \mathbf{x}_i(t)} \right], \Phi_i(t_0, t_0) = I \quad (10)$$

To measure the departure from linearity we adopt, the following ideas, presented originally in reference [4]

$$\nu(t, t_0) \triangleq \sup_{i=1 \dots N} \frac{\|\Phi_i(t, t_0) - \bar{\Phi}(t, t_0)\|}{\|\bar{\Phi}(t, t_0)\|} \quad (11)$$

The nonlinearity index of equation (11) is a deterministic measure associated with a particular domain of initial conditions, and a particular interval of time following prescription of the initial conditions. Alternatively, by taking expectations over distributions of initial conditions and over time, this concept has obvious extensions to associated probabilistic measures [4]. We also mention that the nonlinearity index for dynamical systems defined by equation (11) is related to the Lyapunov exponent [9] which is defined as the average rate of divergence (convergence) of two neighboring trajectories as measured by

$$\lambda_k = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{|\delta \mathbf{x}_k(t)|}{|\delta \mathbf{x}_k(t_0)|} = \lim_{t \rightarrow \infty} \frac{1}{t} \ln |\bar{\Phi}_k(t, t_0)|, k = 1 \dots n \quad (12)$$

Where, $\bar{\Phi}_k$ denotes the k^{th} row of state transition matrix, $\bar{\Phi}$. The number of Lyapunov exponents is equal to the number of dimensions of the phase space (n), but it is common to just refer to the largest one. Therefore, if we define $\bar{\Lambda}$ to be the maximum of all possible Lyapunov exponent then we can write $\bar{\Lambda}$ as

$$\bar{\Lambda} = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \|\bar{\Phi}\|_2 \quad (13)$$

Therefore, the nonlinear index, $\nu(t, t_0)$ can be sought as

$$\lim_{t \rightarrow \infty} \nu(t, t_0) \sim \sup_{i=1 \dots N} \frac{e^{\Lambda_i} - e^{\bar{\Lambda}}}{e^{\bar{\Lambda}}} \quad (14)$$

where Λ_i is evaluated along the i^{th} trajectory i.e. $\mathbf{x}_i$. The main drawback of Lyapunov's exponent is that they are defined for infinite time limit so they do not reflect the rate of divergence (convergence) for short time period, whereas the nonlinear index defined in equation (11) can be used to measure rate of divergence (convergence) for different parts of trajectory.

These ideas can be focused with reference to Fig. 2, for the case of two variables. Note the nonlinearity index has a different value at each point in time, and the time interval ($t - t_0$) over which the motion is considered is important in most problems. The more general circumstance is indicated in Fig. 3. The expected trajectory of a dynamical system, along with the N trajectories ensuing from the N uniformly spaced initial conditions on a worst-case surface are mapped through time, and the time-varying deformation of the cloud of worst-case points is illustrated at a typical time in the motion. In the most general case envisioned, these points need not remain on simply-connected close surfaces, and will not be simply connected on problems having a separatrix or other condition where bifurcations occur in the domain of interest. The surfaces shown are for illustration (and should be expected for the near-linear, small ν case), but is not assumed in these developments. The expected trajectory initiates at the center of the ball of worst-case initial conditions; the location of all points on the neighboring worst-case trajectories predicted as linear departures from the nominal (expected) nonlinear trajectory are

$$\mathbf{x}_i(t) = \bar{\mathbf{x}}(t) + \delta \mathbf{x}_i(t), \delta \mathbf{x}_i(t) \cong \bar{\Phi}(t, t_0) \delta \mathbf{x}_i(t_0) \quad (15)$$

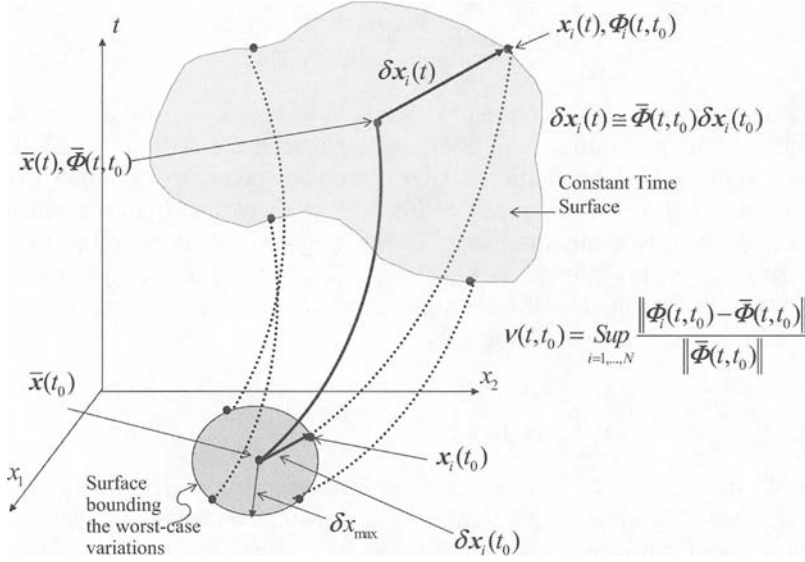


FIG. 2. Worst-Case Initial Condition Variations: A Nonlinear Dynamical System With a Two-Dimensional State Space.

To the extent that linearization is exact, of course, $\Phi_i(t, t_0) = \bar{\Phi}(t, t_0)$ for all points along the trajectories $\mathbf{x}_i(t)$ and $\bar{\mathbf{x}}(t)$ the approximation errors in equation (15) are negligible. Otherwise, the generally nonlinear and arbitrary behavior of the given system would result in significant errors. If the system is perfectly linear in a given vicinity of the expected trajectory, then the state transition matrix evaluated along all neighboring trajectories in the linear region would clearly be exactly equal to the state transition matrix evaluated at the same time along the expected trajectory, and thus we have a qualitative motivation for the nonlinearity measure of equation (11). Further motivation will be provided by studying below the utility of this idea for various coordinate choices in astrodynamics.

While no proof is offered, it has been found that uniformly spacing the points on the worst-case surface leads to dramatically faster convergence of ν than, for

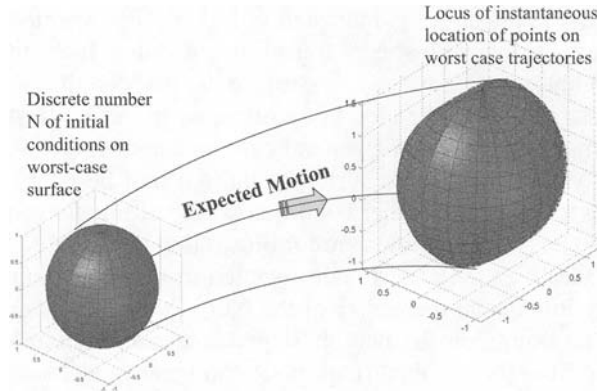


FIG. 3. Nonlinear Propagation of Worst-Case Initial Conditions.

example, using Gaussian Monte Carlo random perturbations on initial conditions to estimate the first two statistical moments of departures from the expected trajectory. The rate of convergence relative to the Monte Carlo method is illustrated in the numerical studies below in the “How Nonlinear Is It” section, where in the examples studied one to two orders of magnitude fewer trials are required for convergence.

For the case of redundant coordinate systems, the above developments require modification. In particular, for the case of quaternion parameterization of rotations, small departure motions are best represented not by additive, but rather by multiplicative perturbations, as we will present below. In particular, we will see that the small errors usually used to justify linear error theory approximations also lead to a nice way to locally eliminate the redundant degree of freedom—as has been exploited by Shuster, Markley, et. al [10] for quaternion-based attitude estimation and associated covariance analysis. Slight modifications of the methodology are required to accommodate multiplicative perturbations as we show below. Generally speaking, care must always be taken to constrain the variations about nominal motions to be consistent with explicit or implicit constraints.

Rotational Kinematics

We begin by considering now the orthogonal projection of a generally rotating orthogonal triad of unit vectors $\{\hat{\mathbf{b}}\}$ relative to a second triad $\{\hat{\mathbf{n}}\}$ fixed in the inertial frame, written as the vectrix [11] equation

$$\{\hat{\mathbf{b}}\} = [C]\{\hat{\mathbf{n}}\} \quad (16)$$

In the event that the unit vector triad $\{\hat{\mathbf{b}}\}$ is moving generally relative to $\{\hat{\mathbf{n}}\}$, with arbitrary angular velocity $\boldsymbol{\omega} = [\omega_1 \ \omega_2 \ \omega_3]^T$, the direction cosine matrix satisfies the (rigorously linear, regular) differential equation [12]

$$[\dot{C}] = -[\tilde{\boldsymbol{\omega}}][C], [\tilde{\boldsymbol{\omega}}] = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix} \quad (17)$$

The $[C]$ matrix is heavily redundant, satisfying the orthogonality condition $[C]^T[C] = [C][C]^T = I$; this orthogonality condition is a rigorous integral of equation (17).

Euler Angles

As is well-known [11–15], all minimum-parameter descriptions of rigid body rotation require three coordinates; the most fundamental issues relating to the choice of attitude coordinates center on the functional relationship to the direction cosine matrix and the behavior of the corresponding kinematic differential equation derivable from equation (17). Perhaps the most frequently used representation of $[C]$ is the 3-1-3 Euler angle parameterization

$$[C(\phi, \theta, \psi)] = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (18)$$

The corresponding kinematic differential equation follows from equation (17) as

$$\begin{Bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{Bmatrix} = \frac{1}{\sin \theta} \begin{bmatrix} \sin \psi & \cos \psi & 0 \\ \cos \psi \sin \theta & -\sin \psi \sin \theta & 0 \\ -\sin \psi \cos \theta & -\cos \psi \cos \theta & \sin \theta \end{bmatrix} \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{Bmatrix} \quad (19)$$

This differential equation is nonlinear and contains singularities at $\theta = 0, \pi, 2\pi, \dots$. The twelve classical Euler angle ($i-j-k$) sets [11–13] have singularities at either odd multiples of π (for the six *symmetric* sets with the first and third indices being the same) or odd multiples of $\pi/2$ (for the six *asymmetric* sets with no repeated indices). We give for subsequent reference the kinematic differential equation for the 3-2-1 (yaw-pitch-roll) Euler angles, which are frequently used in spacecraft and aircraft attitude dynamics:

$$\begin{Bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{Bmatrix} = \frac{1}{\cos \theta} \begin{bmatrix} 0 & \sin \psi & \cos \psi \\ 0 & \cos \psi \cos \theta & -\sin \psi \cos \theta \\ \cos \theta & \sin \psi \sin \theta & \cos \psi \sin \theta \end{bmatrix} \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{Bmatrix} \quad (20)$$

In contrast to the 3-1-3 Euler angle kinematics of equation (19), the 3-2-1 kinematics of equation (20) linearizes well near all three angles equal zero and is singular for $\|\theta\| = \text{odd multiples of } \pi/2$. Geometrically, it can be shown that singularities always occur for the Euler angles whenever the second angle assumes values that cause the axis of rotation for the first and third angles to be colinear.

The most elegant sets of attitude coordinates are associated with *Euler's Principal Rotation* theorem (the most general angular displacement is equivalent to a single rotation ϕ about the eigenvector $\mathbf{l}$ of $[C]$ corresponding to the eigenvalue $+1$). The most fundamental principal rotation representation of $[C]$ is given in terms of ϕ and $\mathbf{l}$ [13]:

$$[C(\mathbf{l}, \phi)] =$$

$$\begin{bmatrix} l_1^2(1 - \cos \phi) + \cos \phi & l_1 l_2(1 - \cos \phi) + l_3 \sin \phi & l_1 l_3(1 - \cos \phi) - l_2 \sin \phi \\ l_2 l_1(1 - \cos \phi) - l_3 \sin \phi & l_2^2(1 - \cos \phi) + \cos \phi & l_2 l_3(1 - \cos \phi) + l_1 \sin \phi \\ l_3 l_1(1 - \cos \phi) + l_2 \sin \phi & l_3 l_2(1 - \cos \phi) - l_1 \sin \phi & l_3^2(1 - \cos \phi) + \cos \phi \end{bmatrix} \quad (21)$$

or compactly

$$[C(\mathbf{l}, \phi)] = \cos \phi [I] + (1 - \cos \phi) [\mathbf{l} \mathbf{l}^T] - \sin \phi \tilde{\mathbf{l}} \quad (22)$$

Making use of the identities: $\cos \phi = 2 \cos^2 \frac{\phi}{2} - 1$, $\sin \phi = 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2}$, equations (21), (22) can be brought to several attractive forms. In particular, consider the following three elegant parameterizations [12, 13, 15].

Quaternion (Euler Parameters)

Following Euler and Hamilton [12, 13, 15], we define the four elements (the *Euler parameters*) of the quaternion as

$$q_i = l_i \sin \frac{\phi}{2}, i = 1, \dots, 3, q_4 = \cos \frac{\phi}{2} \quad (23)$$

As can be verified by inspection, these parameters are once-redundant, locating a point on a four-dimensional unit sphere as

$$\mathbf{q}^T \mathbf{q} = \sum_{i=1}^4 q_i^2 = 1 \quad (24)$$

The direction cosine matrix is then parameterized as a function of the quaternion elements (Euler Parameters) as

$$[C(\mathbf{q})] = \begin{bmatrix} q_1^2 - q_2^2 - q_3^2 + q_4^2 & 2(q_1 q_2 + q_4 q_3) & 2(q_1 q_3 - q_4 q_2) \\ 2(q_1 q_2 - q_4 q_3) & -q_1^2 + q_2^2 - q_3^2 + q_4^2 & 2(q_2 q_3 + q_4 q_1) \\ 2(q_1 q_3 + q_4 q_2) & 2(q_2 q_3 - q_4 q_1) & -q_1^2 - q_2^2 + q_3^2 + q_4^2 \end{bmatrix} \quad (25)$$

The most general dynamical evolution of the direction cosine matrix is thus mapped onto a curve traced out by the quaternion vector $\mathbf{q}(t)$ on the four dimensional unit sphere. In fact, since all elements of $\mathbf{q}$ appear as quadratic products in $[C(\mathbf{q})]$, it is evident that $[C(\mathbf{q})] = [C(-\mathbf{q})]$ is invariant with respect to a sign change of all elements of $\mathbf{q}$; both $\mathbf{q}(t)$ and the image trajectory $-\mathbf{q}(t)$ correspond to the same physical rotational motion.

Substitution of equation (25) into equation (17) leads to the remarkable differential equation

$$\begin{aligned} \dot{\mathbf{q}} &= \frac{1}{2} [\Omega] \mathbf{q}, [\Omega] = \begin{bmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{bmatrix} \\ &= \frac{1}{2} [B(\mathbf{q})] \begin{Bmatrix} \omega_2 \\ \omega_2 \\ \omega_3 \\ 0 \end{Bmatrix}, [B(\mathbf{q})] = \begin{bmatrix} q_4 & -q_3 & q_2 & q_1 \\ q_3 & q_4 & -q_1 & q_2 \\ -q_2 & q_1 & q_4 & q_3 \\ -q_1 & -q_2 & -q_3 & q_4 \end{bmatrix} \end{aligned} \quad (26)$$

Notice that the $[B]$ matrix is orthogonal, and thus the kinematics have at once been regularized and rigorously linearized. It is easy to verify that equation (24) is an exact integral of equation (26). Comparing with equation (20) it is evident that these coordinates have substantial advantages over the Euler angles, especially as regards the simplicity of the governing kinematical differential equation. Equation (26) is linear and regular, without approximation, whereas equation (20) is nonlinear and contains singularities.

Since equation (26) is linear in $\mathbf{q}$, the general solution has the form

$$\mathbf{q}(t) = \Phi(t, t_0) \mathbf{q}(t_0), \quad \dot{\Phi}(t, t_0) = \frac{1}{2} [\Omega] \Phi(t, t_0), \quad \Phi(t_0, t_0) = I \quad (27)$$

This mapping suggests that arbitrary initial condition variations of the quaternion are mapped into the instantaneous quaternion variations as

$$\delta \mathbf{q}(t) = \Phi(t, t_0) \delta \mathbf{q}(t_0) \quad (28)$$

However, variations of initial conditions are not arbitrary; the variations must always be admissible. While the constraint of equation (24) is a rigorous integral of equation (26), admissible initial conditions must satisfy the norm constraint of equation (24) and all admissible variations in initial conditions must satisfy the first variation of equation (24).

Introducing the vector part of the quaternion, $\mathbf{q}$, we write

$$\mathbf{q} = \begin{Bmatrix} \underline{\mathbf{q}} \\ q_4 \end{Bmatrix} \quad (29)$$

Then the norm constraint of equation (24) can be rewritten as

$$\underline{\mathbf{q}}^T \underline{\mathbf{q}} + q_4^2 - 1 = 0 \quad (30)$$

Taking the first variation of equation (24), then it follows that the admissible differential variations $\{\delta q_1, \delta q_2, \delta q_3, \delta q_4\}$ must lie in the tangent plane defined by

$$\sum_{i=1}^3 q_i \delta q_i + q_4 \delta q_4 \equiv \underline{\mathbf{q}}^T \delta \underline{\mathbf{q}} + q_4 \delta q_4 = 0 \quad (31)$$

It is useful to partition $\Phi(t, t_0)$ as

$$\Phi(t, t_0) \equiv \begin{bmatrix} \Phi_{\underline{\mathbf{q}}\underline{\mathbf{q}}} & \Phi_{\underline{\mathbf{q}}q_4} \\ \Phi_{q_4\underline{\mathbf{q}}} & \Phi_{q_4q_4} \end{bmatrix} \quad (32)$$

Now equation (28) expands to reveal

$$\delta \underline{\mathbf{q}}(t) = \Phi(t, t_0) \delta \underline{\mathbf{q}}(t_0) \Rightarrow \begin{cases} \delta \underline{\mathbf{q}}(t) = \Phi_{\underline{\mathbf{q}}\underline{\mathbf{q}}}(t, t_0) \delta \underline{\mathbf{q}}(t_0) + \Phi_{\underline{\mathbf{q}}q_4}(t, t_0) \delta q_4(t_0) \\ \delta q_4(t) = \Phi_{q_4\underline{\mathbf{q}}}(t, t_0) \delta \underline{\mathbf{q}}(t_0) + \Phi_{q_4q_4}(t, t_0) \delta q_4(t_0) \end{cases} \quad (33)$$

and from equation (31), we see the constrained variation of initial conditions must satisfy $\{\underline{\mathbf{q}}(t_0)\}^T \{\delta \underline{\mathbf{q}}(t_0)\} + q_4(t_0) \delta q_4(t_0) = 0$, so from equation (33), we conclude that the vector part of the quaternion maps from initial conditions as

$$\delta \underline{\mathbf{q}}(t) = \left[\Phi_{\underline{\mathbf{q}}\underline{\mathbf{q}}}(t, t_0) - \frac{1}{q_4(t_0)} \Phi_{\underline{\mathbf{q}}q_4}(t, t_0) \{\underline{\mathbf{q}}(t_0)\}^T \right] \delta \underline{\mathbf{q}}(t_0) \equiv [\Phi_c(t, t_0)] \delta \underline{\mathbf{q}}(t_0) \quad (34)$$

Since we are free to choose the inertial frame, we can virtually assure that $q_4(t_0) \approx 1$ and this mapping is always well-behaved. The 3×3 matrix $[\Phi_c(t, t_0)]$ in equation (34) represents the constrained Jacobian partial derivatives of $\mathbf{q}(t)$ with respect to $\mathbf{q}(t_0)$, and generally is full rank. The constrained, locally linearized mapping of equation (34) can be used for covariance analysis, solving two-point boundary problems, and other applications where it is necessary to have a minimal coordinate representation, while at the same time enjoying the globally linear, regular properties of the quaternion kinematical differential equation of equation (26). Notice that paradox: the once-redundant 4-vector $\mathbf{q}(t)$ satisfies a globally linear differential equation, however, minimal coordinate departure motion from any reference trajectory is only approximately linearly described by the minimal coordinate vector $\mathbf{q}(t)$ parameterizing small departure rotations.

The quaternion representation and the results of equations (25), (26) can be derived directly from equations (18), (19) as a coordinate transformation from the 3-1-3 Euler angles [13]

$$\begin{aligned} q_1 &= s \frac{\theta}{2} c \left(\frac{\phi - \psi}{2} \right), \quad q_2 = s \frac{\theta}{2} s \left(\frac{\phi - \psi}{2} \right), \\ q_3 &= c \frac{\theta}{2} s \left(\frac{\phi + \psi}{2} \right), \quad q_4 = c \frac{\theta}{2} c \left(\frac{\phi + \psi}{2} \right) \end{aligned} \quad (35)$$

where “c” and “s” abbreviate cos and sin respectively. Junkins and Turner [13] give the corresponding transformations for all twelve classical Euler angle sets.

The well-known and widely used equation (26) is an elegant regularization and linearization of rotational rigid body rotational kinematics. At the expense of one redundant coordinate, the singularities present in all three parameter sets, such as the divide by zero at $\theta = 0, \pi$ in equation (19), or in equation (20) at $\theta = \pi/2$ are completely eliminated. Note, among all known representations of rotational motion, only two, the direction cosine matrix kinematics of equation (7) and the quaternion kinematic differential equation of equation (26) are rigorously linear, regular and nonsingular. All other parameterizations have nonlinear kinematic differential equations that contain singularities. We now consider two attitude coordinate choices that are closely related to the quaternion.

One of the most important properties of the quaternion is the elegant way successive rotations can be represented as a single composite rotation. Specifically, consider equation (25) with two successive rotations represented by the direction cosine matrix multiplication as

$$[C(\mathbf{q}'')] = [C(\mathbf{q}')] [C(\mathbf{q})] \quad (36)$$

The corresponding quaternion vectors are related by the quaternion multiplication operation

$$\mathbf{q}'' = [M(\mathbf{q}')] \mathbf{q} \quad (37)$$

where the quaternion multiplication operator is the orthogonal matrix

$$[M(\mathbf{q}')] = \begin{bmatrix} q'_4 & q'_3 & -q'_2 & q'_1 \\ -q'_3 & q'_4 & q'_1 & q'_2 \\ q'_2 & -q'_1 & q'_4 & q'_3 \\ -q'_1 & -q'_2 & -q'_3 & q'_4 \end{bmatrix} \quad (38)$$

Rodrigues Parameters (Gibbs Vector)

The Gibbs vector is conveniently defined in terms of the principal rotation vector and principal rotation angle as [12]

$$\mathbf{g} = \mathbf{l} \tan \frac{\phi}{2} \equiv \begin{Bmatrix} q_1/q_4 \\ q_2/q_4 \\ q_3/q_4 \end{Bmatrix} \Rightarrow [C(\mathbf{g})] = \frac{1}{1 + \mathbf{g}^T \mathbf{g}} [(1 - \mathbf{g}^T \mathbf{g})I + \mathbf{g} \mathbf{g}^T - 2\tilde{\mathbf{g}}] \quad (39)$$

The kinematic differential equation follows from equation (17) as

$$\dot{\mathbf{g}} = \frac{1}{2} [I + \tilde{\mathbf{g}} + \mathbf{g} \mathbf{g}^T] \boldsymbol{\omega} \quad (40)$$

While the singular behavior $\|\mathbf{g}\| \rightarrow \infty$ as $\|\phi\| \rightarrow \pi$ as is not evident by inspection of equation (40), it is evident from the definition of equation (39). The singular behavior is readily verified by any solution of equation (40). However for $\|\phi\| \leq \pi/2$, it is easy to verify that equation (40) is very well behaved; the unit sphere $\|\mathbf{g}\| = 1$ corresponds to a rotation of $\|\phi\| = \pi/2$ about all infinity of rotation axes. Note from equation (39) that reversing the sign of all four q_i does not change the unique $\mathbf{q}$ vector; both $\mathbf{q}$ and its image map to same $\mathbf{g}$.

Modified Rodrigues Parameters (MRPs)

The Modified Rodrigues Parameters (MRPs) are defined in terms of the principal rotation vector [12, 14, 16] as

$$\boldsymbol{\sigma} = \mathbf{l} \tan \frac{\phi}{4} \equiv \begin{Bmatrix} q_1/(1 + q_4) \\ q_2/(1 + q_4) \\ q_3/(1 + q_4) \end{Bmatrix} \Rightarrow [C(\boldsymbol{\sigma})] = [I + (1 + \boldsymbol{\sigma}^T \boldsymbol{\sigma})^{-2}(8[\tilde{\boldsymbol{\sigma}}]^2 - 4(1 - \boldsymbol{\sigma}^T \boldsymbol{\sigma})[\tilde{\boldsymbol{\sigma}}])] \quad (41)$$

The MRP kinematic differential equation follows from equation (17) as

$$\dot{\boldsymbol{\sigma}} = \frac{1}{4}[B(\boldsymbol{\sigma})]\boldsymbol{\omega}, [B(\boldsymbol{\sigma})] = [(1 - \boldsymbol{\sigma}^T \boldsymbol{\sigma})I + 2[\tilde{\boldsymbol{\sigma}}] + 2\boldsymbol{\sigma}\boldsymbol{\sigma}^T] \quad (42)$$

$$[B(\boldsymbol{\sigma})]^T[B(\boldsymbol{\sigma})] = [B(\boldsymbol{\sigma})][B(\boldsymbol{\sigma})]^T = (1 + \boldsymbol{\sigma}^T \boldsymbol{\sigma})^2[I] \quad (43)$$

Among all known three parameter descriptions of attitude, only the MRPs have the remarkable property that the kinematic coefficient matrix has orthogonal rows and columns (equation (43)).

While the singular behavior $\|\boldsymbol{\sigma}\| \rightarrow \infty$ as $\|\phi\| \rightarrow 2\pi$ is not evident by inspection of equation (42), it is evident from the definition of equation (41); the singular behavior is readily verified by any solution of equation (42). However for $\|\phi\| \leq \pi$, it is also easy to verify that the MRPs and the solution of equation (42) are extremely well behaved; the unit sphere $\|\boldsymbol{\sigma}\| = 1$ corresponds to all rotations up to and including all possible $\|\phi\| = \pi$ angular displacements. However, a full revolution of the body about any axis is required to encounter the singularity $\|\boldsymbol{\sigma}\| \rightarrow \infty$ as $\|\phi\| \rightarrow 2\pi$, and one half-revolution of the body about any axis always remains within the most regular and near linear region (the unit $\boldsymbol{\sigma}$ -sphere). Thus the parameters are well behaved for most large motions exclusive of tumbling motions.

Note the important property: For each given orientation there are two distinct $\boldsymbol{\sigma}$ -vectors; one always locates a point inside the unit sphere $\|\boldsymbol{\sigma}\| \leq 1$, while the second (the image or shadow point) always locates a point outside the unit sphere $\|\boldsymbol{\sigma}_s\| \geq 1$. The two sets of MRPs correspond to the quaternion $\mathbf{q}$ and its image $-\mathbf{q}$. Upon reversing the sign of the quaternion elements in the definition of equation (41), we obtain the image (shadow) MRP transformation as a function $\boldsymbol{\sigma}$ as

$$\boldsymbol{\sigma}_s = \left(\frac{-1}{\boldsymbol{\sigma}^T \boldsymbol{\sigma}} \right) \boldsymbol{\sigma} \quad (44)$$

Analogous to the quaternion image points, the MRP image points have opposite signs, but unlike the quaternion, the image points have a non-unit scaling. Note when $\boldsymbol{\sigma}$ approaches the origin, the shadow image point $\boldsymbol{\sigma}_s$ approaches infinity and vice versa. Both $\boldsymbol{\sigma}$ and $\boldsymbol{\sigma}_s$ satisfy exactly the same kinematic differential equation (42). Thus, it is possible to switch from any MRP trajectory outside the unit sphere to the corresponding image trajectory inside the unit sphere and never encounter the singularity. The two sets of MRPs are also elegantly interpreted using the stereographic projection [12] from the quaternion unit sphere as shown in Fig. 4. As a consequence of these properties, the Modified Rodrigues Parameters enjoy a special place among all the minimal parameter representations of attitude. Their singularity free state space volume ($\pm 2\pi$ about any axis) is substantially larger than

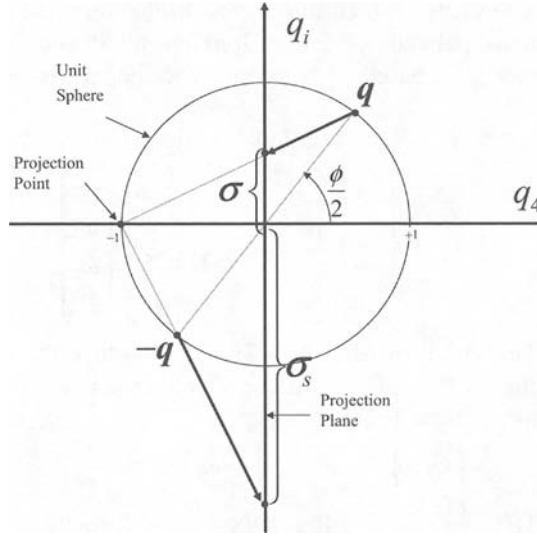


FIG. 4. Stereographic Projection of Quaternion Unit Sphere: Geometric Relationship of Quaternion and MRP Image Points.

that of any Euler angle representation (never farther than $\pi/2$ from a singularity) and the classical RP representation ($\pm\pi$ about any axis). Also, among all three parameter representations, only the MRPs have a kinematic operator $[B(\sigma)] = [(1 - \sigma^T \sigma)I + 2[\tilde{\sigma}] + 2\sigma\sigma^T]$ of equation (42) with orthogonal rows and columns. The numerical studies presented below in the “How Nonlinear Is It?” section are sufficient to suggest that the MRPs are the most nearly linear three-parameter representation of attitude. This completes our discussion of rotation kinematics and we now turn to a unique formulation of rotational dynamics.

Rotational Dynamics

Eulerian Dynamics

First we consider the Eulerian viewpoint. Torque = d/dt (angular momentum) leads naturally to the familiar equations

$$\begin{aligned} L_1 &= I_1 \dot{\omega}_1 - (I_2 - I_3)\omega_2\omega_3 \\ L_2 &= I_2 \dot{\omega}_2 - (I_3 - I_1)\omega_3\omega_1 \\ L_3 &= I_3 \dot{\omega}_3 - (I_1 - I_2)\omega_1\omega_2 \end{aligned} \quad (45)$$

where $\{\omega_1, \omega_2, \omega_3\}$ are the body axis components of angular velocity, $\{L_1, L_2, L_3\}$ are the body axis components of an arbitrary torque, and $\{I_1, I_2, I_3\}$ are the body principal inertias. These three differential equations can be combined with any of the foregoing kinematic differential equations for a particular coordinate choice, for example, if we choose the MRPs, then equation (45) would be solved simultaneously with equation (42), whereas, if we choose the quaternion, then equation (45) would be solved simultaneously with equation (26).

Morton's Canonical Quaternion Formulation

Morton [17] formulated Hamilton's Canonical equations for rotational dynamics, using as generalized coordinates the quaternion, and as momentum coordinates,

the corresponding conjugate momentum vector. In the year just prior to Morton's death, this work was formalized by Morton, Junkins and Shuster into the 1993 journal article of reference [17], based upon a conference paper Morton wrote a decade earlier.

The kinetic energy of rigid body rotational motion is given by

$$2T = I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2 = \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ 0 \end{Bmatrix}^T \begin{bmatrix} I_1 & & & \\ & I_2 & & \\ & & I_3 & \\ & & & I_4 \end{bmatrix} \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ 0 \end{Bmatrix} \triangleq \tilde{\boldsymbol{\omega}}^T [\tilde{I}] \tilde{\boldsymbol{\omega}} \quad (46)$$

where we follow the pattern of Morton [17] and introduce the arbitrary quantity $I_4 = H^2/(2T)$, having units of inertia, where H is the total angular momentum.

The angular momentum vector is

$$\mathbf{H} = \begin{Bmatrix} I_1\omega_1 \\ I_2\omega_2 \\ I_3\omega_3 \\ 0 \end{Bmatrix}, \tilde{\mathbf{H}} = \begin{Bmatrix} I_1\omega_1 \\ I_2\omega_2 \\ I_3\omega_3 \\ 0 \end{Bmatrix} = [\text{diag}(I_i)] \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ 0 \end{Bmatrix} = 2[\text{diag}(I_i)][B(\mathbf{q})]^T \dot{\mathbf{q}} \quad (47)$$

Substituting from equations (26) into equation (46), we are led to the Hamiltonian

$$\mathcal{H} = T = 2\dot{\mathbf{q}}^T [J(\mathbf{q})] \dot{\mathbf{q}} \quad (48)$$

where the generalized inertia matrix is

$$[J(\mathbf{q})] = [B(\mathbf{q})][\text{diag}(I_i)][B(\mathbf{q})]^T \quad (49)$$

and the matrix $[B(\mathbf{q})]$ is given by equation (26).

The generalized momentum vector $\mathbf{p}$ corresponding to the quaternion $\mathbf{q}$ is

$$\mathbf{p} = \frac{\partial T}{\partial \dot{\mathbf{q}}} = 4[B(\mathbf{q})][\text{diag}(I_i)][B(\mathbf{q})]^T \dot{\mathbf{q}}, \Rightarrow \dot{\mathbf{q}} = \frac{1}{4}[B(\mathbf{q})][\text{diag}(1/I_i)][B(\mathbf{q})]^T \mathbf{p} \quad (50)$$

Furthermore, it is evident from equation (47) that

$$\mathbf{p} = \frac{\partial T}{\partial \dot{\mathbf{q}}} = 2[B(\mathbf{q})]\tilde{\mathbf{H}}, \Rightarrow \tilde{\mathbf{H}} = \frac{1}{2}[B(\mathbf{q})]^T \mathbf{p} = -\frac{1}{2}[B(\mathbf{p})]^T \mathbf{q} \quad (51)$$

From this, it follows that

$$p^2 = \mathbf{p}^T \mathbf{p} = 4H^2, \Rightarrow p = 2H \quad (52)$$

Using equation (50) in equation (48), we can express the Hamiltonian in the form

$$\mathcal{H}(\mathbf{q}, \mathbf{p}) = \frac{1}{8} \mathbf{p}^T [B(\mathbf{q})][\text{diag}(1/I_i)][B(\mathbf{q})]^T \mathbf{p} \quad (53)$$

We can make use of the usual Euler equation³

$$\mathbf{L} = \frac{d}{dt}(\mathbf{H})_N = \frac{d}{dt}(\mathbf{H})_B + \boldsymbol{\omega} \times \mathbf{H} \quad (54)$$

³Subscript $_N$ and $_B$ denote vector differentiation in the inertial and body frame respectively.

and, in the augmented space, we can write

$$\begin{Bmatrix} L_1 \\ L_2 \\ L_3 \\ 0 \end{Bmatrix} = \begin{Bmatrix} \dot{H}_1 \\ \dot{H}_2 \\ \dot{H}_3 \\ 0 \end{Bmatrix} + \begin{bmatrix} 0 & -\omega_3 & \omega_2 & 0 \\ \omega_3 & 0 & -\omega_1 & 0 \\ -\omega_2 & \omega_1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} H_1 \\ H_2 \\ H_3 \\ 0 \end{Bmatrix} \triangleq \bar{\mathbf{L}} = \dot{\mathbf{H}} + [\tilde{\boldsymbol{\omega}}]\bar{\mathbf{H}} \quad (55)$$

Substitution from equations (51), (52), and (26), equation (55) ultimately leads to the differential equation

$$\dot{\mathbf{p}} = -\frac{1}{4}[B(\mathbf{p})][\text{diag}(1/I_i)][B(\mathbf{p})]^T \mathbf{q} + 2[B(\mathbf{q})]\bar{\mathbf{L}} \quad (56)$$

Alternatively and more efficiently, the equations of motion follow from Hamilton's canonical equations by differentiation of the Hamiltonian of equation (52) as

$$\dot{\mathbf{q}} = \frac{\partial \mathcal{H}}{\partial \mathbf{p}} = \frac{1}{4}[B(\mathbf{q})][\text{diag}(1/I_i)][B(\mathbf{q})]^T \mathbf{p} \quad (57)$$

and

$$\dot{\mathbf{p}} = -\frac{\partial \mathcal{H}}{\partial \mathbf{q}} + \mathbf{Q} = -\frac{1}{4}[B(\mathbf{p})][\text{diag}(1/I_i)][B(\mathbf{p})]^T \mathbf{q} + 2[B(\mathbf{q})]\bar{\mathbf{L}} \quad (58)$$

where the generalized moment is $\mathbf{Q} = 2[B(\mathbf{q})]\bar{\mathbf{L}}$. Equations (57) and (58) verify the previous results of equations (50) and (56). Look carefully at the symmetry in equations (57) and (58). There are two remarkable orthogonality properties of their solution:

$$\dot{\mathbf{q}}^T \mathbf{q} = 0 \text{ and } \mathbf{q}^T \mathbf{p} = 0 \quad (59)$$

The first of these is true in general for the quaternion description of kinematics; it follows immediately from differentiation of $\mathbf{q}^T \mathbf{q} = 1$. The second property $\mathbf{q}^T \mathbf{p} = 0$ follows from expanding equation (51) as

$$\begin{aligned} p_1 &= 2(q_4 H_1 - q_3 H_2 + q_2 H_3), & p_2 &= 2(q_3 H_1 + q_4 H_2 - q_1 H_3) \\ p_3 &= 2(-q_2 H_1 + q_1 H_2 + q_4 H_3), & p_4 &= 2(-q_1 H_1 - q_2 H_2 - q_3 H_3) \end{aligned}$$

then the orthogonality property $\mathbf{q}^T \mathbf{p} = \sum_{i=1}^4 q_i p_i = 0$ is verified by direct calculation. Since both $\dot{\mathbf{q}}(t)$ and $\mathbf{p}(t)$ are orthogonal to the instantaneous quaternion $\mathbf{q}(t)$, then they lie in a hyperplane that is always tangent to the unit sphere $\mathbf{q}^T \mathbf{q} = 1$ at $\mathbf{q}(t)$. Furthermore, investigating the total energy we find the elegant result

$$2T = \boldsymbol{\omega}^T \mathbf{H} = \tilde{\boldsymbol{\omega}}^T \bar{\mathbf{H}} = \mathbf{p}^T \dot{\mathbf{q}} \quad (60)$$

In the special case of torque-free motion, notice that Hamilton's canonical equations, equations (57), (58) can be rewritten as

$$\dot{\mathbf{q}} = \frac{1}{4}[B(\mathbf{q})][\text{diag}(1/I_i)][B(\mathbf{q})]^T \mathbf{p} = \frac{1}{2}[B(\mathbf{q})][\text{diag}(1/I_i)]\bar{\mathbf{H}} \quad (61)$$

$$\dot{\mathbf{p}} = -\frac{1}{4}[B(\mathbf{p})][\text{diag}(1/I_i)][B(\mathbf{p})]^T \mathbf{q} = \frac{1}{2}[B(\mathbf{p})][\text{diag}(1/I_i)]\bar{\mathbf{H}} \quad (62)$$

Making use of the fact that $\bar{\omega} = [\omega_1 \ \omega_2 \ \omega_3 \ 0]^T = [\text{diag}(1/I_i)]\bar{\mathbf{H}}$, and referring to equation (26), then equations (50), (51) simplify to the remarkable results

$$\dot{\mathbf{q}} = \frac{1}{2} \begin{bmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{bmatrix} \mathbf{q} = \frac{1}{2} \begin{bmatrix} q_4 & -q_3 & q_2 & q_1 \\ q_3 & q_4 & -q_1 & q_2 \\ -q_2 & q_1 & q_4 & q_3 \\ -q_1 & -q_2 & -q_3 & q_4 \end{bmatrix} \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ 0 \end{Bmatrix} = \frac{1}{2} [B(\mathbf{q})] \bar{\omega} \quad (63)$$

$$\dot{\mathbf{p}} = \frac{1}{2} \begin{bmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{bmatrix} \mathbf{p} = \frac{1}{2} \begin{bmatrix} p_4 & -p_3 & p_2 & p_1 \\ p_3 & p_4 & -p_1 & p_2 \\ -p_2 & p_1 & p_4 & p_3 \\ -p_1 & -p_2 & -p_3 & p_4 \end{bmatrix} \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ 0 \end{Bmatrix} = \frac{1}{2} [B(\mathbf{p})] \bar{\omega} \quad (64)$$

As a consequence of these governing differential equations, we conclude that *both* $\mathbf{q}$ and $\mathbf{p}$ are quaternions! Notice, however, $\mathbf{p}$ is not a unit quaternion, but rather has constant magnitude $2H$, as is evident from equation (65). Since $\omega(t)$ is common between equations (63), (64); the solutions for $\mathbf{q}(t)$ and $\mathbf{p}(t)$ *can differ only as a consequence of their different initial conditions*. Further reflection on this truth reveals that the unit quaternion $\mathbf{p}(t)/(2H)$ describes exactly the same attitude history as does $\mathbf{q}(t)$, however, there is a constant rotation between the two, established by initial condition $\mathbf{p}(t_0)$. Note $\mathbf{p}(t_0)/(2H)$ defines the orientation of a second inertial frame associated with the constant angular momentum vector. Linearization of Morton's formulation with respect to initial conditions introduces certain subtleties. Because it is a twice redundant formulation; two constraints must be imposed. In essence, the quaternion $\mathbf{q}(t)$, $\mathbf{q}(t_0)$, must be constrained to have a unit norm so the norm variation must be set to zero, whereas the conjugate momentum's variation $\mathbf{p}(t)$, $\mathbf{p}(t_0)$, must be constrained so that the attitude quaternion vector is orthogonal to the conjugate momentum vector i.e. $\mathbf{q}^T \mathbf{p} = 0$. We present these equations below in the "How Nonlinear Is It?" section.

In summary, Morton's formulation leads to the symmetrical differential equations (57), (58) for the general case, and to the extremely elegant equations (63), (64) for the torque-free case. These bring remarkable symmetry to the kinematics and kinetics of rigid body motion. Finally, we note that Deprit and Elipse [18] used Morton's canonical variables $(\mathbf{q}, \mathbf{p})$ as the starting point for subsequent canonical transformations that reduce rigid body motion to an elegant action/angle form and establishes connections to the Serret-Andoyer canonical variables [19].

Classical and Regularized Orbital Dynamics

Cartesian Nonrotating Coordinates Centered in the Primary Body

The classical differential equations governing the perturbed two-body problem (masses m_1 and m_2 , universal gravity constant G) are the familiar Cartesian coordinate differential equations [2]

$$\begin{aligned} \ddot{x} + \frac{\mu x}{r^3} &= a_{dx}(t, x, y, z, \dot{x}, \dot{y}, \dot{z}), \quad r^2 = x^2 + y^2 + z^2 \\ \ddot{y} + \frac{\mu y}{r^3} &= a_{dy}(t, x, y, z, \dot{x}, \dot{y}, \dot{z}), \quad (\cdot) \equiv d(\cdot)/dt \\ \ddot{z} + \frac{\mu z}{r^3} &= a_{dz}(t, x, y, z, \dot{x}, \dot{y}, \dot{z}), \quad \mu \equiv G(m_1 + m_2) \end{aligned} \quad (65)$$

or

$$\ddot{\mathbf{r}} + \frac{\mu \mathbf{r}}{r^3} = \mathbf{a}_d \quad (66)$$

Given the initial conditions $\{\mathbf{r}(t_0), \dot{\mathbf{r}}(t_0)\}$, and prescription of the perturbing acceleration function $\mathbf{a}_d$, equations (65), (66) can be solved for the Cartesian coordinate trajectory. Notice the singularity at $\mathbf{r} \rightarrow 0$; while all finite celestial bodies preclude exact encounters of the singular condition, the presence of this singularity is still of interest in causing variably nonlinear behavior for small $\mathbf{r}$. Of course these equations are analytically solvable for Keplerian motion, in the event that we neglect the perturbing acceleration on the right-hand sides of equations (65). For virtually all other cases, we require either a numerical differential equation solver (Cowell's Method) or a perturbation/expansion approach to establish analytical approximations.

There are several known transformations of both dependent and independent coordinates that regularize (eliminate the singularity at $\mathbf{r} \rightarrow 0$) the nonlinear differential equations of equations (65), (66), for the special case of zero perturbations. In the developments below, we summarize some of the most important of these developments.

Orbital Dynamics Regularization via the Burdet Transformation

An appealing coordinate transformation was developed by Burdet [20, 21]. His choice of dependent coordinates are $\{\hat{\mathbf{x}}, \hat{\mathbf{x}}', \rho, \rho', p\}$ and the independent variable he selected is the true anomaly ϕ , where transformation from $\{\mathbf{r}, \dot{\mathbf{r}}, t\}$ to $\{\hat{\mathbf{x}}, \hat{\mathbf{x}}', \rho, \rho', p, \phi\}$ is

$$\begin{aligned} \hat{\mathbf{x}} &= \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \mathbf{r}/r \\ \rho &= 1/r \\ \mu p^* &= r^2(\dot{\mathbf{r}}^T \dot{\mathbf{r}} - \dot{r}^2) \equiv \|\mathbf{r} \times \dot{\mathbf{r}}\|^2 \\ dt &= \frac{1}{\sqrt{\mu p}} r^2 d\phi \Rightarrow (\cdot)' \triangleq \frac{d}{d\phi}(\cdot) = \left(\frac{r^2}{\sqrt{\mu p}} \right) \frac{d}{dt}(\cdot) \end{aligned} \quad (67)$$

Taking the indicated $(\cdot)'$ derivatives, equation (67) map the classical differential equations (66) into the equations

$$\left. \begin{aligned} \hat{\mathbf{x}}'' + \hat{\mathbf{x}} &= \left(\frac{1}{\mu p \rho^3} \right) [\mathbf{a}_d - (\hat{\mathbf{x}} \cdot \mathbf{a}_d) \hat{\mathbf{x}}] - \frac{1}{2} \frac{p'}{p} \hat{\mathbf{x}}' \\ \rho'' + \rho &= \frac{1}{p} \left(1 - \frac{1}{2} \rho' p' + \frac{1}{\mu \rho^2} (\hat{\mathbf{x}} \cdot \mathbf{a}_d) \right) \\ p' &= \frac{2}{\mu \rho^3} (\hat{\mathbf{x}}' \cdot \mathbf{a}_d) \end{aligned} \right\} \quad (68)$$

In the absence of perturbations, these equations are regular and linear (exclusive of the time equation), and so Keplerian motion simplifies to the following linear differential equations, with the analytical solutions indicated:

$$\begin{aligned}
\hat{\mathbf{x}}'' + \hat{\mathbf{x}} = 0 &\Rightarrow \left\{ \begin{aligned} \hat{\mathbf{x}} &= \frac{\mathbf{r}}{r} = \hat{\mathbf{x}}_0 \cos \phi + \hat{\mathbf{y}}_0 \sin \phi, \phi = f - f_0 \\ \hat{\mathbf{y}} &= \frac{d\hat{\mathbf{x}}}{d\phi} = -\hat{\mathbf{x}}_0 \sin \phi + \hat{\mathbf{y}}_0 \cos \phi \end{aligned} \right. \\
\rho'' + \rho = \frac{1}{p} &\Rightarrow \left\{ \begin{aligned} \rho &= \frac{1}{r} = \frac{1}{p} (1 + e \cos \phi) \\ \rho' &= -(e \sin \phi)/p \end{aligned} \right. \\
p' = 0 &\Rightarrow p = \frac{r_0^2}{\mu} (\dot{\mathbf{r}}_0 \cdot \dot{\mathbf{r}}_0 - \dot{\mathbf{r}}_0^2) \\
dt = \frac{1}{\sqrt{\mu p}} r^2 d\phi &\Rightarrow \left\{ \begin{aligned} t - t_0 &= \frac{[p/(1 - e^2)]^{3/2}}{\sqrt{\mu}} [\phi + G(\phi) - G(0)] \\ G(\phi) &= 2 \tan^{-1} \left(\frac{p\rho'}{p\rho + \sqrt{1 - e^2}} \right) + \frac{\sqrt{1 - e^2}}{\rho} \rho' \end{aligned} \right.
\end{aligned} \quad (69)$$

The inverse transformation to obtain $\{\mathbf{r}(t), \dot{\mathbf{r}}(t)\}$ is obvious by inspection.

Note the unperturbed problem has no small divisors except as we approach any rectilinear orbit for which $p \rightarrow 0$, and $e \rightarrow 1$. Notice for the general case of equations (68), including the time equation, is a 10th order system, compared to the original 6th order system of equations (66). Thus regularization comes at the expense of increased dimensionality. While the transformed differential equations for the dependent state variables are linear for Keplerian motion, the time equation (as is to be expected) is nonlinear (a transformed cousin of Kepler's equation). Furthermore, note that while the Keplerian motion appears linear and uncoupled, the equations are actually not independent, in the sense that they all depend in a coupled way on independently specified initial conditions through the nonlinear dependence of the parameters p and e on initial conditions.

Regularization via the KS Transformation

A remarkable regularization of the planar two-body problem was discovered in 1906 by Levi-Civita [22]; the generalization of this regularization to the general problem went unsolved until the 1965 dissertation of Kustaanheimo [23] who developed this result in collaboration with Stiefel. Subsequently, Stiefel and Scheiffle published the important monograph *Linear and Regular Celestial Mechanics* [24], which brought the KS approach to a mature state. Subsequent studies were addressed to a variety of specific celestial mechanics problems and applications [25–28]. An observant reader will notice some analogies to the quaternion representation of rigid body kinematics.

The Kustaanheimo-Stiefel (KS) transformation from the position vector components is given by

$$\bar{\mathbf{r}} = \begin{Bmatrix} x \\ y \\ z \\ 0 \end{Bmatrix} = \begin{bmatrix} u_1 & -u_2 & -u_3 & u_4 \\ u_2 & u_1 & -u_4 & -u_3 \\ u_3 & u_4 & u_1 & u_2 \\ u_4 & -u_3 & u_2 & -u_1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} \Leftrightarrow \begin{cases} x = u_1^2 - u_2^2 - u_3^2 + u_4^2 \\ y = 2(u_1 u_2 - u_3 u_4) \\ z = 2(u_1 u_3 + u_2 u_4) \end{cases} \quad (70)$$

or, compactly

$$\bar{\mathbf{r}} = [L(\mathbf{u})]\mathbf{u} \quad (71)$$

The 4×4 matrix operator $[L(\mathbf{u})]$ has orthogonal rows and columns and reminds us of quaternion multiplication and the operator $[B(\mathbf{q})]$ in equation (26). The inverse of $[L(\mathbf{u})]$ is

$$[L(\mathbf{u})]^{-1} = \frac{1}{r} [L(\mathbf{u})]^T \quad (72)$$

and

$$r = \sum_{i=1}^4 u_i^2 = \mathbf{u}^T \mathbf{u} \quad (73)$$

The transformation from (x, y, z) to $\mathbf{u}$ is not unique, but combining equation (73) with equation (70), it is easy to verify the traditional solution

$$\begin{aligned} x \geq 0: \quad u_1 &= \sqrt{\frac{r+x}{2}}, \quad u_2 = \frac{y}{2u_1}, \quad u_3 = \frac{z}{2u_1}, \quad u_4 = 0 \\ x < 0: \quad u_1 &= \frac{y}{2u_2}, \quad u_2 = \sqrt{\frac{r-x}{2}}, \quad u_3 = 0, \quad u_4 = \frac{z}{2u_2} \end{aligned} \quad (74)$$

The solution of equation (74) is branched in order to make the divisions to be as far from zero as possible. Note that the above transformation is required only to *initialize* the KS variables; the transformation from $\mathbf{u}$ to $\mathbf{r}$ is *always unique and non-singular*, given by equations (70).

In addition to the dependent coordinate transformation of equations (70), we utilize for the independent (time) variable the universal eccentric anomaly s , having the differential relationship to (t, r) as

$$dt = r ds \Rightarrow \frac{d(\cdot)}{dt} = \frac{1}{r} (\cdot)', \quad \frac{d^2(\cdot)}{dt^2} = \frac{1}{r^3} [r(\cdot)'' - r'(\cdot)'], \quad (\cdot)' \triangleq \frac{d(\cdot)}{ds} \quad (75)$$

The transformations of equations (67) and (75) applied to the perturbed two-body problem in the form

$$\begin{aligned} \frac{d^2x}{dt^2} + \frac{\mu x}{r^3} &= a_{dx} = a_{dx}^* - \frac{\partial V}{\partial x} \\ \frac{d^2y}{dt^2} + \frac{\mu y}{r^3} &= a_{dy} = a_{dy}^* - \frac{\partial V}{\partial y} \\ \frac{d^2z}{dt^2} + \frac{\mu z}{r^3} &= a_{dz} = a_{dz}^* - \frac{\partial V}{\partial z} \end{aligned} \quad (76)$$

lead to the perturbed linear oscillator in $\mathbf{u}$ -space

$$\mathbf{u}'' + \alpha \mathbf{u} = \mathbf{Q} \quad (77)$$

where

$$\alpha' = -\mathbf{u}'^T [L(\mathbf{u})]^T \mathbf{a}_d^* - \frac{r}{2} \frac{\partial V}{\partial t} \quad (78)$$

$$\alpha = -\frac{1}{2} [\text{total energy}] = -\frac{1}{2} \left[\frac{\dot{\mathbf{r}} \cdot \dot{\mathbf{r}}}{2} + \left(-\frac{\mu}{r} + V \right) \right] = \frac{1}{r} \left[\frac{\mu}{2} - \mathbf{u}'^T \mathbf{u}' \right] - \frac{V}{2} \quad (79)$$

$$\mathbf{Q} = \frac{r}{2} [L(\mathbf{u})]^T \mathbf{a}_d - \frac{1}{4} \frac{\partial(rV)}{\partial \mathbf{u}} \quad (80)$$

and $V(x, y, x, t)$ is the potential energy of the conservative part of the perturbing acceleration. In the absence of nonconservative perturbations ($\mathbf{a}_d^* = 0$), it is evident from equation (78) that α is constant unless V depends explicitly on t .

Remarkably, for unperturbed Keplerian motion with the generalized force $\mathbf{Q} = 0$, the classical differential equations are transformed into four rigorously linear and uncoupled second-order differential equations, and motion near the singularity at $r = 0$ of the Cartesian differential equations (67) is replaced by regular motion in the transformed space, equation (77), at the nonsingular equilibrium point $\mathbf{u} = 0$. All of the difficulties have been absorbed into the time equation and the elegant change in dependent variables. The transformation and the resulting differential equations (65) hold for all energies and geometries of orbits, including hyperbolic and parabolic cases; the unperturbed motion can be solved universally in terms of either the complex exponential functions or with the real universal functions developed by Herrick, Battin and Stumpff, see references [2, 23, 29]. Analogous to the Burdet formulation, these apparently linear uncoupled differential equations are not independent in the sense that they depend (through the parameter α) on the independent initial conditions.

Analogy Between Orbit and Attitude Dynamics

Osculating Frame Kinematics

For a general orbital motion, consider the osculating triad $T\{\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}\}$ of unit vectors

$$\hat{\mathbf{x}} = \mathbf{r}/r, \quad \hat{\mathbf{z}} = (\mathbf{r} \times \dot{\mathbf{r}})/\|\mathbf{r} \times \dot{\mathbf{r}}\|, \quad \hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{x}} \quad (81)$$

With reference to Fig. 5, we can think of this frame as being imbedded in a fictitious rigid body—this suggests using rotational dynamics concepts to describe the rotational motion of this frame. These ideas were explored in references [30, 31]. To capture the essence of this analogy, first let the instantaneous inertial angular velocity of this frame be denoted by $\boldsymbol{\omega}$. Taking components of vectors along the radial, tangential and normal unit vectors, we write

$$\begin{aligned} \mathbf{r} &= r\hat{\mathbf{x}} \\ \mathbf{a}_d &= a_{dx}\hat{\mathbf{x}} + a_{dy}\hat{\mathbf{y}} + a_{dz}\hat{\mathbf{z}} \\ \boldsymbol{\omega} &= \omega_x\hat{\mathbf{x}} + \omega_y\hat{\mathbf{y}} + \omega_z\hat{\mathbf{z}} \end{aligned} \quad (82)$$

The velocity vector generally has radial, tangential and normal components given by

$$\dot{\mathbf{r}} = \dot{r}\hat{\mathbf{x}} + \boldsymbol{\omega} \times (r\hat{\mathbf{x}}) = \dot{r}\hat{\mathbf{x}} + r\omega_z\hat{\mathbf{y}} - r\omega_y\hat{\mathbf{z}} \quad (83)$$

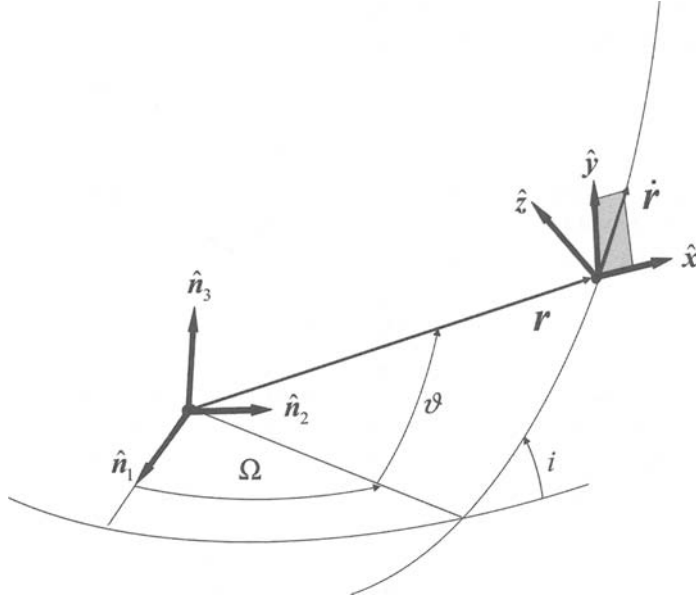


FIG. 5. Osculating Triad Geometry.

However, we require the osculating frame to be compatible with the instantaneous osculating two-body orbit. Since the osculating orbit lies in the plane defined by $(\mathbf{r}, \dot{\mathbf{r}})$, there is zero velocity normal to the orbit plane, thus we conclude from equation (83) that the tangential component of angular velocity of the osculating frame must vanish ($\omega_y = 0$) as an orbit osculation constraint.

Recall for particle motion that the angular momentum per unit mass is

$$\mathbf{h} = \mathbf{r} \times \dot{\mathbf{r}} = r^2(\omega_y \hat{\mathbf{y}} + \omega_z \hat{\mathbf{z}}) \Rightarrow \frac{d}{dt}(\mathbf{h})_N = \mathbf{r} \times \mathbf{a}_d = (0)\hat{\mathbf{x}} + (-ra_{dz})\hat{\mathbf{y}} + (ra_{dy})\hat{\mathbf{z}} \quad (84)$$

Also, note a second expression for the inertial time derivative of angular momentum can be obtained from

$$\dot{\mathbf{h}} = \frac{d}{dt}(\mathbf{h})_N = \frac{d}{dt}(\mathbf{h})_T + \boldsymbol{\omega} \times \mathbf{h} \quad (85)$$

Upon substitution of equations (82)–(84) into equation (85), and equating to the second of equations (84), we are led to “Euler’s” moment equations for orbital motion:

$$\begin{aligned} \text{radial:} & \quad 0 = 0 \\ \text{transverse:} & \quad \dot{\omega}_y = \omega_x \omega_z - a_{dz}/r - 2(\dot{r}/r)\omega_y \\ \text{normal:} & \quad \dot{\omega}_z = -\omega_x \omega_y + a_{dy}/r - 2(\dot{r}/r)\omega_z \end{aligned} \quad (86)$$

Comparing these with Euler’s equations for rigid body motion, namely, equations (45), we see that the radial equation is trivial, and the transverse and normal equations contain “variable inertia terms,” involving $(r, \dot{r})$, that are not present in

the rigid body version of Euler's equations. These equations are immediately simplified further upon imposing the osculation constraint ($\omega_y = 0$) to the equations

$$\begin{aligned} \text{transverse:} \quad & 0 = \omega_x \omega_z - a_{dz}/r \\ \text{normal:} \quad & \dot{\omega}_z = a_{dy}/r - 2(\dot{r}/r)\omega_z \end{aligned} \quad (87)$$

Thus the transverse equation is thus reduced to an *algebraic gyroscopic condition* from which we can immediately solve for the orbit precession angular rate to be the radial component of angular velocity

$$\omega_x = \frac{a_{dz}}{r\omega_z} \quad (88)$$

Thus the orbit “rolls” about the instantaneous radius vector due to the presence of the perturbing acceleration a_{dz} normal to the orbit plane. This is a nice path to an elegant interpretation of orbital precession. Looking now at the normal equation of equations (87), for the case of $a_{dy} = 0$, this can be rearranged to verify Keplers 2nd law as

$$\frac{\dot{\omega}_z}{\omega_z} = -2 \frac{\dot{r}}{r} \Rightarrow \ln(\omega_z) = -2 \ln(r) + \text{constant} \Rightarrow r^2 \omega_z = \text{constant} \quad (89)$$

Now, taking the analogy further, we know the kinematics for 3-1-3 Euler angles (Ω, i, ϑ) are given by equation (19) with the correspondence $(\phi, \theta, \psi) \rightarrow (\Omega, i, \vartheta)$, so the governing kinematic differential equation is

$$\begin{Bmatrix} \frac{d\Omega}{dt} \\ \frac{di}{dt} \\ \frac{d\vartheta}{dt} \end{Bmatrix} = \frac{1}{\sin i} \begin{bmatrix} \sin \vartheta & \cos \vartheta & 0 \\ \cos \vartheta \sin i & -\sin \vartheta \sin i & 0 \\ -\sin \vartheta \cos i & -\cos \vartheta \cos i & \sin i \end{bmatrix} \begin{Bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{Bmatrix}, \quad \vartheta = \omega + f \quad (90)$$

Imposing the osculation constraint $\omega_y = 0$ and making use of equation (88), we immediately obtain from equation (90) the familiar variation of parameters differential equations for the longitude of the ascending node, the inclination, and the reduced latitude as

$$\frac{d\Omega}{dt} = \left(\frac{\sin \vartheta}{\sin i} \right) \left(\frac{a_{dz}}{r\omega_z} \right), \quad \frac{di}{dt} = \cos \vartheta \left(\frac{a_{dz}}{r\omega_z} \right), \quad \frac{d\vartheta}{dt} = \omega_z - \left(\frac{\sin \vartheta}{\tan i} \right) \left(\frac{a_{dz}}{r\omega_z} \right) \quad (91)$$

So, it is evident that the rigid body analogy is useful, both in terms of qualitative insights and in efficient derivations. Moreover, we can bring forward all we know about regularization of rotational kinematics and directly make good use of these ideas in the orbit problem. In particular, the singularities associated with zero inclination can be readily eliminated and the rotational kinematic differential equations completely regularized—if we simply make a judicious choice of orientation coordinates to define the angular position of the osculating triad.

Quaternion Regularization of the Osculating Triad Kinematics

Upon choosing the quaternion to parameterize the inertial orientation of the osculating triad, we can immediately replace the classical 3-1-3 kinematics of equations (90)–(91) by the quaternion kinematical equation of equation (26), specialized ($\omega_y = 0$) here as

$$\dot{\mathbf{q}} = \frac{1}{2} \begin{bmatrix} 0 & \omega_z & -\omega_y & \omega_x \\ -\omega_z & 0 & \omega_x & \omega_y \\ \omega_y & -\omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix} \mathbf{q} = \frac{1}{2} \begin{bmatrix} q_4 & -q_3 & q_2 & q_1 \\ q_3 & q_4 & -q_1 & q_2 \\ -q_2 & q_1 & q_4 & q_3 \\ -q_1 & -q_2 & -q_3 & q_4 \end{bmatrix} \begin{Bmatrix} \omega \\ 0 \end{Bmatrix} \quad (92)$$

Furthermore, for Keplerian motion ($\omega_x = 0$), equations (91) decouple into two subsets as

$$\dot{q}_1 = \frac{\omega_z}{2} q_2, \quad \dot{q}_2 = -\frac{\omega_z}{2} q_1, \quad \text{and} \quad \dot{q}_3 = \frac{\omega_z}{2} q_4, \quad \dot{q}_4 = -\frac{\omega_z}{2} q_3 \quad (93)$$

Defining the change in true anomaly as

$$d\phi = \omega_z dt, \text{ or } \phi = \int_{t_0}^t \omega_z d\tau \quad (94)$$

Then the Keplerian solution of equations (93) for the elements of $\mathbf{q}$ is

$$q_i = a_i \cos \frac{\phi}{2} + b_i \sin \frac{\phi}{2}, \quad i = 1, \dots, 4 \quad (95)$$

where the constants (a_i, b_i) are established from initial conditions.

We note that $\mathbf{q}(t)$ establishes the orientation of the osculating triad, but does not provide the complete solution. However, the kinematics of equation (92) can be integrated simultaneously with additional state variables to provide the complete solution.

Quaternion Regularization of Orbital Dynamics

Using the ideas introduced in references [30, 31], we recall $\mathbf{r} = r\hat{\mathbf{x}} = \hat{\mathbf{x}}/\rho$, $\rho \equiv 1/r$, and since

$$\hat{\mathbf{x}} = [C_{11} \ C_{12} \ C_{13}]^T \quad (96)$$

and we substitute the elements from the first row of $[C]$ from equation (25) and write

$$\begin{aligned} x &= r(q_1^2 - q_2^2 - q_3^2 + q_4^2) = (q_1^2 - q_2^2 - q_3^2 + q_4^2)/\rho \\ y &= 2r(q_1 q_2 + q_4 q_3) = 2(q_1 q_2 + q_4 q_3)/\rho \\ z &= 2r(q_1 q_3 - q_4 q_2) = 2(q_1 q_3 - q_4 q_2)/\rho \end{aligned} \quad (97)$$

Also, we consider the semi-latus rectum $p^* = r^2(\dot{\mathbf{r}} \cdot \dot{\mathbf{r}} - \dot{r}^2)/\mu$, and define the perturbed quantity

$$p = p^* + 2r^2 V/\mu = r^2(\dot{\mathbf{r}} \cdot \dot{\mathbf{r}} - \dot{r}^2 + 2V)/\mu \quad (98)$$

Finally, the perturbed true anomaly, ϕ , is introduced through the time transformation

$$dt = \frac{d\phi}{\sqrt{\mu p}} \frac{1}{\rho^2} \quad (99)$$

Applying the coordinate changes of equations (92), (97)–(99), equations (76) are brought to the elegant regularized form [30, 31]

$$\frac{d}{d\phi} \begin{Bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{Bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & \omega_z & 0 & \omega_x \\ -\omega_z & 0 & \omega_x & 0 \\ 0 & -\omega_x & 0 & \omega_z \\ -\omega_x & 0 & -\omega_z & 0 \end{bmatrix} \begin{Bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{Bmatrix} \quad (100)$$

$$\frac{d^2\rho}{d\phi^2} + \rho = \frac{1}{p} + \frac{1}{\mu p \rho^2} [a_{dx}^* - 2\rho V] - \frac{1}{2p} \frac{d\rho}{d\phi} \frac{d\rho}{d\phi} \quad (101)$$

$$\frac{dp}{d\phi} = \frac{2\omega_z}{\mu\rho^3} a_{dy}^* - \frac{2}{\mu\rho^3} \frac{d\rho}{d\phi} (2V + V_r) + \frac{2V_r}{\mu\sqrt{\mu p \rho^4}} \quad (102)$$

$$V_r = \frac{\partial V}{\partial t}, \quad V_r = x \frac{\partial V}{\partial x} + y \frac{\partial V}{\partial y} + z \frac{\partial V}{\partial z} \quad (103)$$

$$\omega_x = \frac{a_{dz}^*}{\mu p \rho^3 \omega_z}, \quad \omega_z = \sqrt{1 - \frac{2V}{\mu p \rho^3}} \quad (104)$$

The independent variable in equation (100) is ϕ , not t , as in equation (92), and the definitions of (ω_x, ω_z) are scaled accordingly in equations (104). For Keplerian motion, equations (96) are rigorously linear and regular, except for the case of rectilinear orbits. For both perturbed and Keplerian orbits, these are fast variables. Including the time equation of equation (99), the system of equations (96) is of eighth order, as compared to tenth order for the KS formulation. In some problems, the method of variation of parameters is preferred. Variation of parameters approaches closely related to the above development and to the Burdet regularization are presented in references [32, 33].

How Nonlinear Is It?

In this section, we present numerical studies of the nonlinearity of the attitude kinematics, attitude dynamics, and orbital dynamics using the classical and regularized formulations presented above. The results are discussed in the three subsets: Attitude Kinematics, Attitude Dynamics, and Orbital Dynamics. In each subsection four formulations will be considered to model the identical physical motion. We place particular emphasis upon including one classical nonlinear formulation contrasting the classical singular differential equations with one or more coordinate choices having regular nonsingular differential equations. For each formulation and the corresponding numerical solutions ensuing from a family of worst-case initial conditions, we use the nonlinearity index to evaluate the extent to which the large departure motion from the expected reference motion can be approximated linearly.

Attitude Kinematics

We consider the kinematics for four attitude representations:

- 3-2-1 Euler Angles Representation (EA321)
- Classical Rodrigues Parameter Representation (RP)
- Modified Rodrigues Parameter Representation (MRP)
- Quaternion Representation (Euler Parameters)

We summarize the kinematic differential equations for each of the four representations and the Jacobian governing evolution of the state transition matrix. Note the first three are minimal parameter descriptions with singular behavior at some orientations as discussed earlier. The quaternion representation is once-redundant

and globally linear and nonsingular. Following the summary of the kinematic equations are the corresponding Jacobians required to compute the state transition matrix needed for the dynamic nonlinearity index. We consider specific numerical simulations with worst-case initial condition variations and study the behavior of the nonlinearity index.

3-2-1 Euler Angles Representation

Kinematic Differential equation:

$$\dot{\Theta} = \begin{Bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{Bmatrix} = \frac{1}{c\theta} \begin{bmatrix} 0 & s\psi & c\psi \\ 0 & c\theta c\psi & -c\theta s\psi \\ c\theta & s\theta s\psi & s\theta c\psi \end{bmatrix} \begin{Bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{Bmatrix} \quad (105)$$

Linear Departure Motion Differential Equations:

$$\delta\dot{\Theta} = \frac{1}{c\theta} \begin{bmatrix} 0 & (s\psi\omega_2 + c\psi\omega_3) \tan \theta & c\psi\omega_2 - s\psi\omega_3 \\ 0 & 0 & -s\psi\omega_2 - c\psi\omega_3 \\ 0 & \frac{s\psi\omega_2 + c\psi\omega_3}{c\theta} & (c\psi\omega_2 - s\psi\omega_3)s\theta \end{bmatrix} \delta\Theta \quad (106)$$

or

$$\delta\dot{\Theta} = A(t)\delta\Theta, \quad \dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0) \quad (107)$$

Rodrigues Parameters (RP)

Kinematic Differential Equation:

$$\dot{\mathbf{g}} = \frac{1}{2}(\mathbf{I} + \tilde{\mathbf{g}} + \mathbf{g}\mathbf{g}^T)\boldsymbol{\omega} \quad (108)$$

Linear Departure Motion differential Equation:

$$\delta\dot{\mathbf{g}} = \frac{1}{2}[\mathbf{g}\boldsymbol{\omega}^T + (\boldsymbol{\omega}^T\mathbf{g})\mathbf{I} - \tilde{\boldsymbol{\omega}}]_{\mathbf{g}}\delta\mathbf{g} \quad (109)$$

or

$$\delta\dot{\mathbf{g}} = A(t)\delta\mathbf{g}, \quad \dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0) \quad (110)$$

Modified Rodrigues Parameters (MRP)

Kinematic Differential Equations:

$$\dot{\boldsymbol{\sigma}} = \frac{1}{4}[(1 - \boldsymbol{\sigma}^T\boldsymbol{\sigma})\mathbf{I} + 2\tilde{\boldsymbol{\sigma}} + 2\boldsymbol{\sigma}\boldsymbol{\sigma}^T]\boldsymbol{\omega} \quad (111)$$

Linear Departure Motion Differential Equations:

$$\delta\dot{\boldsymbol{\sigma}} = \frac{1}{2}[(\boldsymbol{\omega}^T\boldsymbol{\sigma})\mathbf{I} - \boldsymbol{\omega}\boldsymbol{\sigma}^T + \boldsymbol{\sigma}\boldsymbol{\omega}^T - \tilde{\boldsymbol{\omega}}]_{\boldsymbol{\sigma}}\delta\boldsymbol{\sigma} \quad (112)$$

or

$$\delta\dot{\boldsymbol{\sigma}} = A(t)\delta\boldsymbol{\sigma}, \quad \dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0) \quad (113)$$

Quaternion (Euler Parameter) Representation

Kinematic Differential Equations (rigorously linear and regular):

$$\dot{\mathbf{q}} = \frac{1}{2}[\Omega]\mathbf{q}, \quad [\Omega] = \begin{bmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{bmatrix} \quad (114)$$

$$\dot{\Phi}(t, t_0) = [\Omega]\Phi(t, t_0) \quad (115)$$

From equations (29)–(34), the constrained variation of the vector part $\underline{\mathbf{q}}(t)$ of the quaternion $\mathbf{q}(t)$ satisfies the linearized state transition mapping

$$\delta\underline{\mathbf{q}}(t) = \Phi_c(t, t_0)\delta\underline{\mathbf{q}}(t_0) \quad (116)$$

where

$$\Phi_c(t, t_0) \equiv \Phi_{\underline{\mathbf{q}}\underline{\mathbf{q}}}(t, t_0) - \frac{1}{q_4(t_0)}\Phi_{\underline{\mathbf{q}}q_4}(t, t_0)\{\underline{\mathbf{q}}(t_0)\}^T, \quad \Phi(t, t_0) \equiv \begin{bmatrix} \Phi_{q_4} & \Phi_{q_4q_4} \\ \Phi_{q_4\underline{\mathbf{q}}} & \Phi_{q_4\underline{\mathbf{q}}q_4} \end{bmatrix} \quad (117)$$

The variation $\delta\underline{\mathbf{q}}(t)$ of $\underline{\mathbf{q}}(t)$ parameterizes a small rotation from the attitude defined by $\underline{\mathbf{q}}(t)$. We mention, for the special case of the quaternion representations, that the nonlinearity index is concerned with the worst-case variation of $\Phi_c(t, t_0)$, not $\Phi(t, t_0)$. Note the paradox: the full quaternion is rigorously linear, satisfying $\dot{\mathbf{q}}(t) = \Phi(t, t_0)\dot{\mathbf{q}}(t_0)$, however admissible variations in the quaternion must be subject to the unit norm constraint, so the minimal coordinate vector part of the quaternion behaves only approximately linearly due to the nonlinear constraint on admissible variations (these variations must lie in the local tangent plane of the unit sphere $\mathbf{q}^T\mathbf{q} = 1$).

Uniform Distribution of Points on an N-Dimensional Sphere

The worst-case initial conditions for the simulations below are assumed to lie on a closed worst-case surface (we use the 3- σ ellipsoid, generally in the appropriate n -dimensional state space). The problem of having a given N number of points distributed uniformly (“equally spaced”) on a surface in 3- d space has been investigated in [3, 5, 6], with the objective of sampling a surface whose shape varies with time. The problem is generally not solvable exactly, but as the number of points is increased, it has been found that nearly equally spaced points on a given surface can be closely approached using an approximate optimization process. This is intuitively reasonable—as the number of points approaches infinity it is evident that a uniform density distribution should be approached. The method proposed is based on the minimization of a potential function obtained by considering the points as identical attracting (e.g. gravitational) particles constrained to move on the surface. This physical analog leads to an equilibrium configuration that is associated with a minimum energy state, constraining the particles to lie on the surface in the presence of fictitious forces of attraction or repulsion. The same motivating idea has been used in this work, but applied to a unit sphere and generalized for an n -dimensional space (also known as the Thomson Problem [3]). At $t = 0$ N points are distributed randomly on the unit sphere. The potential energy E_0 , associated with a gravitational

type of fictitious potential of the form $V = \frac{1}{2} \sum_{i,j,i \neq j}^N \frac{1}{r_{ij}}$ is evaluated for this distribution, then the total force $\mathbf{F} = -\nabla V$ acting on each particle is computed and the position $\mathbf{r}_i$ of each point is accordingly displaced for a small time interval Δt , taking the constraint $r_i = |\mathbf{r}_i| = 1$ into consideration. The dynamics need not be tracked rigorously, rather the direction of the resulting forces projected tangent to the surface provides a direction for a discrete step (a one parameter search down these directions locates a minimum—the process is repeated recursively). This simple gradient-like minimization process, motivated by the methods of steepest descent, is initiated from random initial conditions, and is iterated until an approximate minimum for the potential energy V is reached. Converged statistics of inter particle distance (mean and standard deviation) provide a stopping criterion; it is usually sufficient that the standard deviation be reduced to less than one percent of the mean. Care has to be taken to remove all the “rigid body freedoms” since this problem obviously has $n(n-1)/2$ cyclic coordinates. This is easily done for $n = 3$ by choosing Point 1 to lie at unit distance on the x -axis and Point 2 to lie in the x - y plane at unit distance. The converged data for each N are saved, so that any subsequent simulations for the same N will not require the minimization process to be repeated. Analogous generalized logic applies to higher dimensioned and non-spherical surfaces [3, 5, 6].

Numerical Study of the Nonlinearity Index for Various Kinematic Representations

The nominal angular motion for simulation purposes is established by prescribing an exact nominal Euler 3-2-1 angle motion; exact angular velocity and exact coordinates for all other representations can then be derived by coordinate transformation. The motion is defined carefully, so that no singularities are encountered in any of the representations. While the exact motion is known, we use a 4th order Runge-Kutta method to solve each set of differential equations. We choose the nominal Euler angle motion

$$\begin{aligned}\bar{\phi}(t) &= \sin 3t \cos 5t, \quad \bar{\theta}(t) = 1.1 \sin 5t, \\ \bar{\psi}(t) &= 0.5 \sin 5t \{0.1 + \sin 3t\}^3, \quad 0 \leq t \leq 25 \text{ sec}\end{aligned}\tag{118}$$

For this judicious choice of angular motion, the maximum Euler principal rotation angle is also always less than $\pi/2$ and therefore neither the classical Rodrigues parameters (singular at a principal rotation of $\phi = \pi$) nor the modified Rodrigues parameters (singular at $\phi = 2\pi$) encounter a singularity. The small secular term and large amplitude of the motion ensures frequent excursions into highly nonlinear regions of the Euler angle representation. Obviously, the nominal motion has an amplitude of well over a radian in the first two coordinates and involves several frequencies higher than might be expected in spacecraft rigid body motion, so this motion could be considered as a relatively general case, but still short of a tumbling motion.

Figures 6 and 7 show the time histories of nominal Euler angles and the corresponding MRP vector. To consider the effect of various attitude parameters on the validity of linear error propagation, N worst-case initial errors are chosen uniformly on a sphere $\delta\phi_0^2 + \delta\theta_0^2 + \delta\psi_0^2 = \left(\frac{10\pi}{180}\right)^2$. The Euler angle initial conditions are mapped consistently through the appropriate coordinate transformation to generate corresponding initial condition variations for the alternative representations.

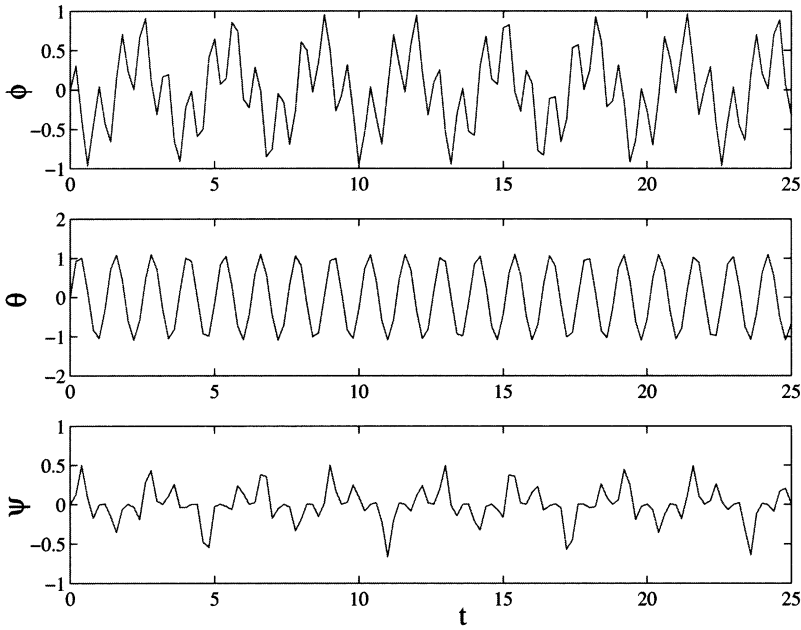


FIG. 6. Nominal Euler 3-2-1 Angles.

Referring to Fig. 6, as a consequence of the large and oscillatory motion, we can expect significant departures from the linear approximations. After some experimentation, the number N of extreme initial conditions was judiciously chosen to ensure that the nonlinearity index $\nu(t, t_0)$ was converged. Figures 8 and 9 show the variation of maximum nonlinear index with sample size N . From Fig. 8, it is clear that $N = 200$ is more than sufficient for the uniformly distributed points on the worst-case

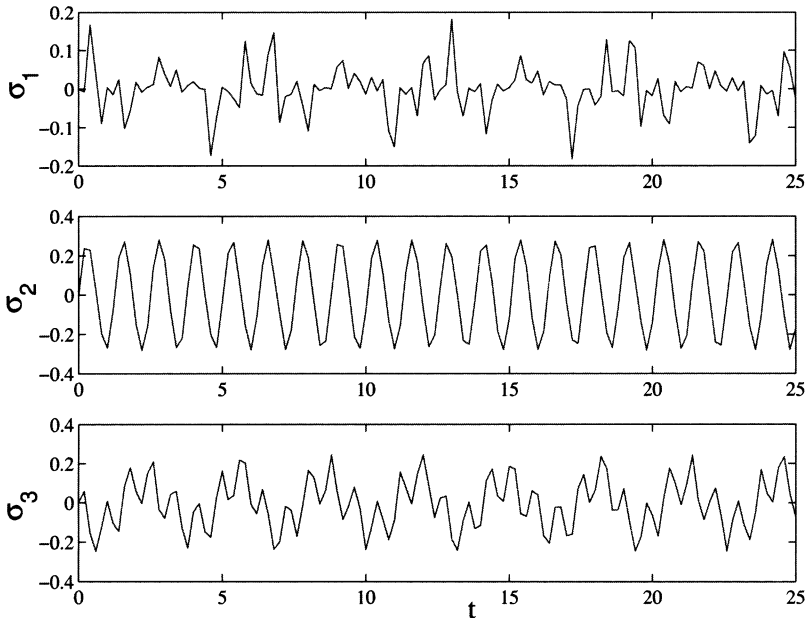


FIG. 7. Nominal Modified Rodrigues Parameters.

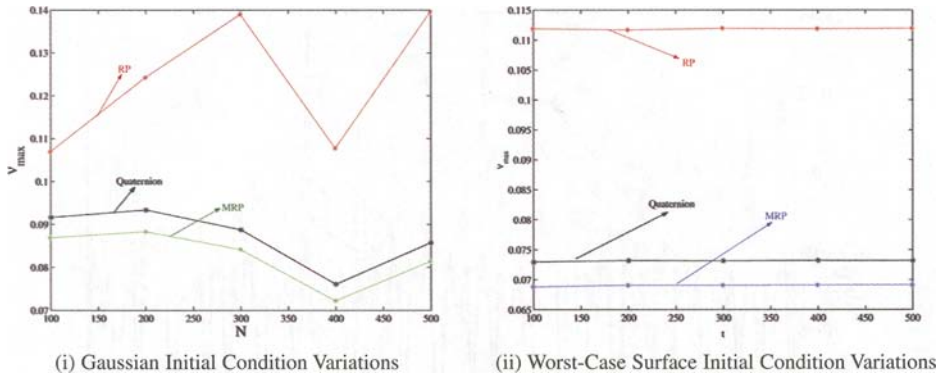


FIG. 8. Maximum Nonlinearity Index $v(t, t_0)$ vs N .

surface, whereas, convergence still is not evident for $N = 500$ for the Gaussian initial condition variations. The poor convergence in the case of Monte Carlo simulation is also due to the fact that the set of 100 initial conditions is independent of the set of 500 initial conditions (i.e. we use independent 100, 200, ..., 500 samples, rather than just accumulate one large sample). We mention that the maximum value of $v(t, t_0)$ found in the Monte Carlo analysis occasionally exceeds that in the worst-case initial conditions; this is because all points in the latter case are constrained to lie on the $3\text{-}\sigma$ surface, whereas, occasionally greater than $3\text{-}\sigma$ outliers arise in the Monte Carlo analysis.

A key issue that needs some attention, when studying nonlinearity measures in various coordinate systems: how do we unify these measures in a “fair” way? This issue is addressed below for the case of attitude kinematics.

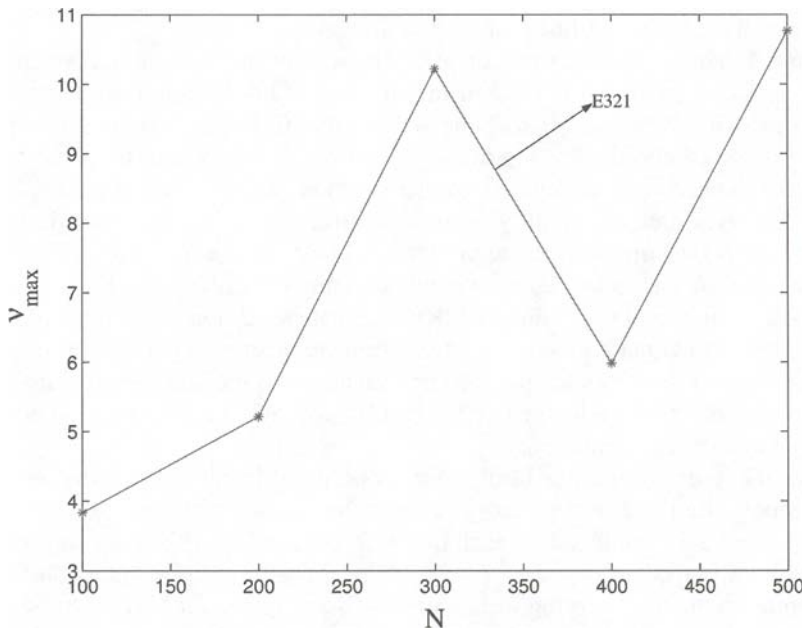


FIG. 9. Maximum Nonlinearity Index $v(t, t_0)$ vs N for Euler Angles.

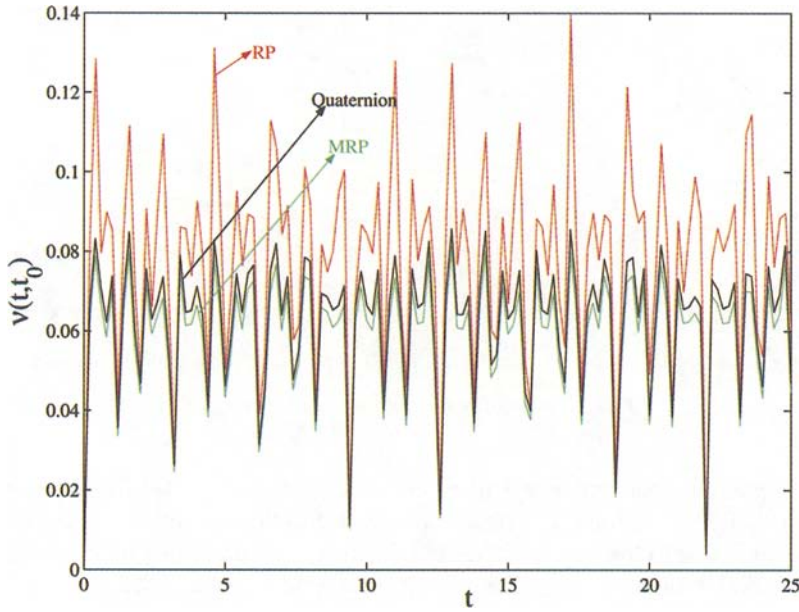


FIG. 10. Nonlinearity Index Variation with Time for Attitude Kinematics.

Figure 10 shows the variation of dynamic nonlinearity index for all attitude parameterizations with time. The computations of the dynamic nonlinearity index are based upon the 500 worst-case initial conditions. As anticipated, the nonlinearity index is $O(> 1)$ for the Euler angles and much smaller for other attitude parameters. From Fig. 10, it is clear that MRP and Quaternion parameterizations have the smallest linear approximation errors ($< 8\%$) for different attitude parameter representations, whereas RP have linear approximation errors of order 12% and Euler angle sets have the worst linear approximation errors $> 100\%$.

Figure 11 shows the linearization error for various attitude parameterizations in their respective spaces. It is evident that the actual linearization errors are in full agreement with the small $\nu(t, t_0)$ behavior in Fig. 10. In Fig. 12, the linear prediction errors of all coordinates are mapped nonlinearly for comparison into the attitude error matrix and compared to the attitude matrix corresponding to zero prediction error, i.e., the identity rotation matrix. Measuring the errors in the attitude matrix is the correct measure of “which is best.” It is evident from Fig. 12 that the attitude errors are much larger when linearizing the Euler 3-2-1 kinematics than when linearizing the Quaternion or MRP kinematics. Although the nonlinearity errors for MRP and Quaternion parameterization are of same order in their respective spaces, when these errors are mapped nonlinearly into the attitude error matrix the Quaternion emerges as winner. The MRP still have lowest approximation errors for three parameter representations.

To gain appreciation of the utility of the validity of linear error theory, in Fig. 13 we compare the linear error theory propagation of the initial worst-case surface (taken to be the $3\text{-}\sigma$ ellipsoid) to each instant in the motion. This is accomplished by the usual covariance propagation⁴ $P(t) = \Phi(t, t_0)P(t_0)\Phi(t, t_0)^T$ and then transforming each point on the time varying instantaneous $3\text{-}\sigma$ ellipsoid $\delta \mathbf{x}^T [P(t)]^{-1} \delta \mathbf{x} = 3^2$ into

⁴ $P(t_0)$ is consistent with the worst-case initial conditions lying on the corresponding $3\text{-}\sigma$ surface.

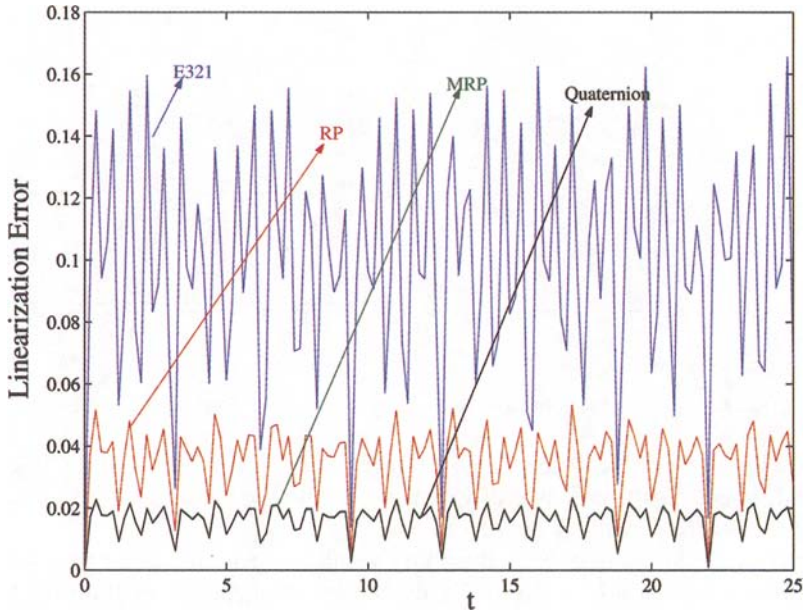


FIG. 11. Average Norm of Attitude Linearization Errors in Respective Attitude Representation Space.

a common coordinate system (we choose the Euler 3-2-1 angles). A Monte Carlo simulation of 100 points is displayed in Fig. 13 superimposed on the linear error theory predicted 3σ ellipsoid projected into two orthogonal planes. As is evident the linearization using the Euler 3-2-1 angle representation is least well correlated to the nonlinear Monte Carlo distribution of points, whereas the Quaternion and Modified Rodrigues Parameters are best correlated to the nonlinear simulation. A

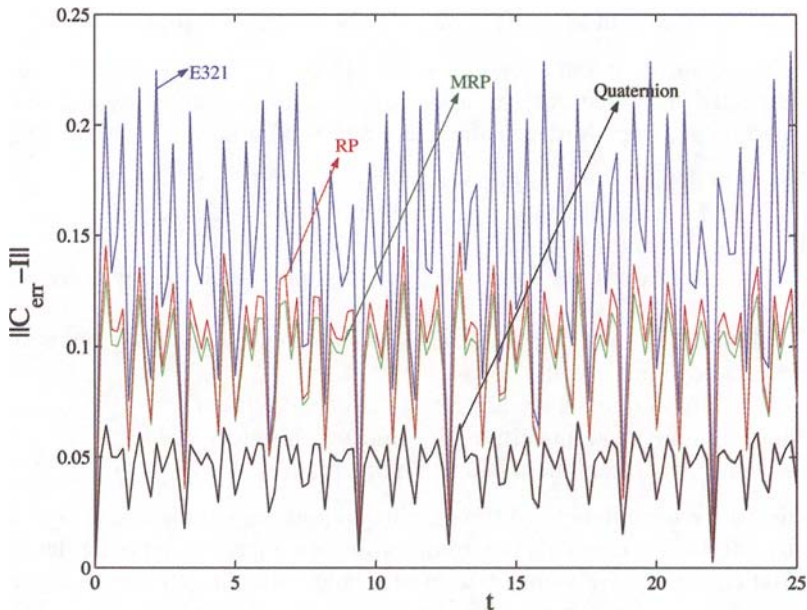


FIG. 12. Average Norm of Attitude Linear Prediction Error for Various Attitude Representations.

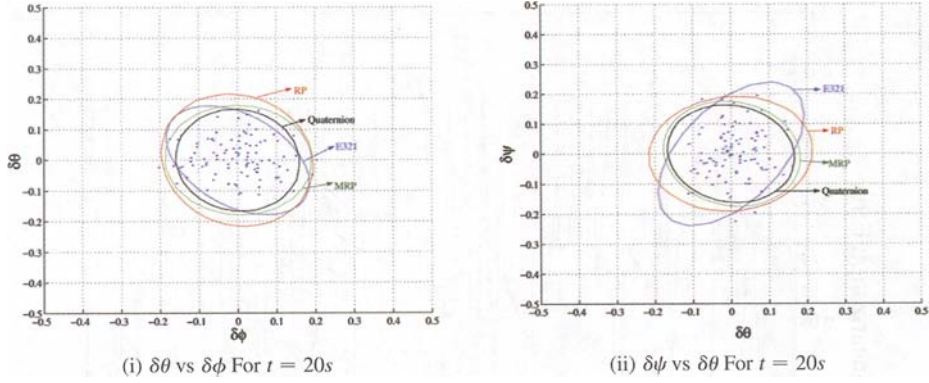


FIG. 13. 3- σ Covariance Surfaces Predicted by Linear Theory in Four Representations.

formal calculation of the statistical covariance matrix is consistent with this graphical trend. These results are also fully consistent with the predictions using the non-linearity index as shown in Fig. 10.

Finally, we mention that while the above results are for one nominal motion and set of worst-case variations, we feel they convey the general situation for typical non-tumbling motions—namely that for large motions, linearization of Euler angles are less accurate than the modified Rodrigues parameters and the quaternion representations.

Attitude Dynamics

For the following prescribed motion, we differentiate the angular velocity coordinates of equation (119) to derive the corresponding acceleration, and with knowledge of the equations of motion and inertias ($I_1 = 100$, $I_2 = 200$, $I_3 = 250 \text{ kg-m}^2$), we determine the torques so that the complete expected/nominal solution is known in advance.

$$\bar{\omega}_1 = 0.01 \sin 5t + 10^{-4}, \quad \bar{\omega}_2 = 0.09 \sin 3t, \quad \bar{\omega}_3 = 0.05 \sin 5t$$

Then, using a differential equation solver (4th order Runge-Kutta) we can solve for the expected and worst-case variations of the motion numerically and be certain of the solution accuracy for five parameterizations of attitude dynamics, namely

- Eulerian dynamics with state variables
—3-2-1 Euler angles and angular velocity along body axes
- Eulerian dynamics with state variables
—Classical Rodrigues Parameters and angular velocity along body axes
- Eulerian dynamics with state variables
—Modified Rodrigues Parameters and angular velocity along body axes
- Eulerian dynamics with state variables
—Quaternion $\mathbf{q}$ and angular velocity along body axes
- Morton's canonical quaternion dynamics with state variables
—Quaternion $\mathbf{q}$ and the conjugate momentum $\mathbf{p}$

The first three formulations' differential equations are nonlinear and contain singularities. Morton's canonical quaternion formulation has regular (singularity-free) differential equations. We summarize briefly below the differential equations governing the state and the state transition matrix for each of the above four coordinate representations of the same physical motion, with Table 1 showing the Jacobian

TABLE 1. Jacobian for Eulerian Dynamical Formulations

Parameters	$\frac{\partial f(\boldsymbol{\sigma}, \boldsymbol{\omega})}{\partial \boldsymbol{\sigma}}$	$\frac{\partial f(\boldsymbol{\sigma}, \boldsymbol{\omega})}{\partial \boldsymbol{\omega}}$
Euler 3-2-1 Angle Set	$\frac{1}{c\theta} \begin{bmatrix} 0 & (s\psi\omega_2 + c\psi\omega_3) \tan \theta & c\psi\omega_2 - s\psi\omega_3 \\ 0 & 0 & -s\psi\omega_2 - c\psi\omega_3 \\ 0 & \frac{s\psi\omega_2 + c\psi\omega_3}{c\theta} & (c\psi\omega_2 - s\psi\omega_3)s\theta \end{bmatrix}$	$\frac{1}{c\theta} \begin{bmatrix} 0 & s\psi & c\psi \\ 0 & c\theta c\psi & -c\theta s\psi \\ c\theta & s\theta s\psi & s\theta c\psi \end{bmatrix}$
RP	$\frac{1}{2}(\mathbf{g}\boldsymbol{\omega}^T + (\boldsymbol{\omega}^T \mathbf{g})I - \tilde{\boldsymbol{\omega}}) _{\mathbf{g}}$	$\frac{1}{2}(I + \tilde{\mathbf{g}} + \mathbf{g}\mathbf{g}^T)$
MRP	$\frac{1}{2}((\boldsymbol{\omega}^T \boldsymbol{\sigma})I - \boldsymbol{\omega}\boldsymbol{\sigma}^T + \boldsymbol{\sigma}\boldsymbol{\omega}^T - \tilde{\boldsymbol{\omega}}) _{\tilde{\boldsymbol{\sigma}}}$	$\frac{1}{4}((1 - \boldsymbol{\sigma}^T \boldsymbol{\sigma})I + 2\tilde{\boldsymbol{\sigma}} + 2\boldsymbol{\sigma}\boldsymbol{\sigma}^T)$
Quaternion	$\frac{1}{2}[\Omega(\boldsymbol{\omega})]$	$\begin{bmatrix} q_4 & -q_3 & q_2 \\ q_3 & q_4 & -q_1 \\ -q_2 & q_1 & q_4 \\ -q_1 & -q_2 & -q_3 \end{bmatrix}$

matrix in each case. Of course, the key underlying issue is evaluation of the validity of linearization about the expected trajectory over the family of large initial condition variations for each of the four coordinate representations.

Eulerian Dynamics:

Differential Equations for State Evolution:

$$\dot{\mathbf{x}} = \begin{Bmatrix} \dot{\boldsymbol{\sigma}} \\ \dot{\boldsymbol{\omega}} \end{Bmatrix} = \begin{Bmatrix} f(\boldsymbol{\sigma}, \boldsymbol{\omega}) \\ I^{-1}(L - \tilde{\boldsymbol{\omega}}I\boldsymbol{\omega}) \end{Bmatrix} \quad (119)$$

Linear Departure Motion Differential Equations:

$$\delta \dot{\mathbf{x}} = \begin{bmatrix} \frac{\partial f(\boldsymbol{\sigma}, \boldsymbol{\omega})}{\partial \boldsymbol{\sigma}} & \frac{\partial f(\boldsymbol{\sigma}, \boldsymbol{\omega})}{\partial \boldsymbol{\omega}} \\ O_{3 \times 3} & I^{-1}(I\tilde{\boldsymbol{\omega}} - \tilde{\boldsymbol{\omega}}I) \end{bmatrix} \delta \mathbf{x} \triangleq A(t)\delta \mathbf{x}, \quad \dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0) \quad (120)$$

Morton's Canonical Quaternion Formulation

Differential Equations for State Evolution:

$$\dot{\mathbf{x}} = \begin{Bmatrix} \dot{\mathbf{q}} \\ \dot{\mathbf{p}} \end{Bmatrix} = \begin{Bmatrix} \frac{1}{4}B(\mathbf{q})[\text{diag}(1/I_i)]B(\mathbf{q})^T \mathbf{p} \\ -\frac{1}{4}B(\mathbf{p})[\text{diag}(1/I_i)]B(\mathbf{p})^T \mathbf{q} + 2B(\mathbf{q})\bar{L} \end{Bmatrix} \quad (121)$$

Linear Departure Motion Differential Equations:

$$\delta \dot{\mathbf{x}} = \begin{bmatrix} \frac{1}{4}(R(NB(\mathbf{q})^T \mathbf{p}) - B(\mathbf{q})NB(\mathbf{p})^T) & \frac{1}{4}(B(\mathbf{p})NB(\mathbf{p})^T) \\ \frac{1}{4}(2R(\bar{L}) - B(\mathbf{p})NB(\mathbf{p})^T) & \frac{1}{4}(R(NB(\mathbf{p})^T \mathbf{q}) - B(\mathbf{p})NB(\mathbf{q})^T) \end{bmatrix} \delta \mathbf{x} \quad (122)$$

$$\delta \dot{\mathbf{x}} = A(t)\delta \mathbf{x}, \quad \dot{\Phi}(t, t_0) = A(t)\Phi(t, t_0) \quad (123)$$

where

$$B(\mathbf{a}) = \begin{bmatrix} a_4 & -a_3 & a_2 & a_1 \\ a_3 & a_4 & -a_1 & a_2 \\ -a_2 & a_1 & a_4 & a_3 \\ -a_1 & -a_2 & -a_3 & a_4 \end{bmatrix}, \quad R(\mathbf{a}) = \begin{bmatrix} a_4 & a_3 & -a_2 & a_1 \\ -a_3 & a_4 & a_1 & a_2 \\ a_2 & -a_1 & a_4 & a_3 \\ -a_1 & -a_2 & -a_3 & a_4 \end{bmatrix} \text{ and}$$

$$N = [\text{diag}(1/I_i)] \quad (124)$$

Equations (122) and (123) suggest that arbitrary initial condition variations of the quaternion and generalized momentum coordinates are mapped into instantaneous quaternion and generalized momentum variations as

$$\begin{Bmatrix} \delta \mathbf{q}(t) \\ \delta \mathbf{p}(t) \end{Bmatrix} = \Phi(t, t_0) \begin{Bmatrix} \delta \mathbf{q}(t_0) \\ \delta \mathbf{p}(t_0) \end{Bmatrix} \quad (125)$$

However, variations of initial conditions are not arbitrary; the variations must always be admissible; i.e., they must satisfy the first variation of equations (24) and (59).

Introducing the first three components of the quaternion and the generalized momentum as $\underline{\mathbf{q}}$ and $\underline{\mathbf{p}}$ respectively, we write

$$\mathbf{q} = \begin{Bmatrix} \underline{\mathbf{q}} \\ q_4 \end{Bmatrix}, \quad \mathbf{p} = \begin{Bmatrix} \underline{\mathbf{p}} \\ p_4 \end{Bmatrix} \quad (126)$$

Now, the norm constraint of equation (24) and orthogonality constraint of equation (59) can be rewritten as

$$\underline{\mathbf{q}}^T \underline{\mathbf{q}} + q_4^2 - 1 = 0 \quad (127)$$

$$\underline{\mathbf{q}}^T \underline{\mathbf{p}} + q_4 p_4 = 0 \quad (128)$$

Taking the first variations of equations (127) and (128), then it follows that the admissible differential variations $\{\delta \underline{\mathbf{q}} \quad \delta q_4 \quad \delta \underline{\mathbf{p}} \quad \delta p_4\}^T$ must lie in the tangent plane defined by

$$\underline{\mathbf{q}}^T \delta \underline{\mathbf{q}} + q_4 \delta q_4 = 0 \quad (129)$$

$$\underline{\mathbf{q}}^T \delta \underline{\mathbf{p}} + \underline{\mathbf{p}}^T \delta \underline{\mathbf{q}} + q_4 \delta p_4 + p_4 \delta q_4 = 0 \quad (130)$$

Solving equations (129) and (130) simultaneously for δp_4 , we obtain

$$\delta p_4 = -\frac{\underline{\mathbf{q}}^T}{q_4} \delta \underline{\mathbf{p}} + \left(\frac{p_4}{q_4^2} \underline{\mathbf{q}}^T - \frac{\underline{\mathbf{p}}^T}{q_4} \right) \delta \underline{\mathbf{q}} \quad (131)$$

It is useful to partition $\Phi(t, t_0)$ as

$$\Phi(t, t_0) = \begin{bmatrix} \Phi_{\mathbf{q}\mathbf{q}} & \Phi_{\mathbf{q}q_4} & \Phi_{\mathbf{q}\mathbf{p}} & \Phi_{\mathbf{q}p_4} \\ \Phi_{q_4\mathbf{q}} & \Phi_{q_4q_4} & \Phi_{q_4\mathbf{p}} & \Phi_{q_4p_4} \\ \Phi_{\mathbf{p}\mathbf{q}} & \Phi_{\mathbf{p}q_4} & \Phi_{\mathbf{p}\mathbf{p}} & \Phi_{\mathbf{p}p_4} \\ \Phi_{p_4\mathbf{q}} & \Phi_{p_4q_4} & \Phi_{p_4\mathbf{p}} & \Phi_{p_4p_4} \end{bmatrix} \quad (132)$$

Now, expanding equation (125) and substituting for $\delta q_4(t_0)$ and $\delta p_4(t_0)$ from equations (129) and (131) respectively, we obtain

$$\Phi_c(t, t_0) = \begin{bmatrix} \Phi_{\underline{q}\underline{q}} - \Phi_{\underline{q}q_4} \frac{\underline{\mathbf{q}}^T(t_0)}{q_4(t_0)} + \Phi_{\underline{q}p_4} \frac{p_4(t_0)\underline{\mathbf{q}}^T(t_0)}{q_4^2(t_0)} - \Phi_{\underline{q}p_4} \frac{\underline{\mathbf{p}}^T(t_0)}{q_4(t_0)} & \Phi_{\underline{q}\underline{\mathbf{p}}} - \Phi_{\underline{q}p_4} \frac{\underline{\mathbf{q}}^T(t_0)}{q_4(t_0)} \\ \Phi_{\underline{\mathbf{p}}\underline{\mathbf{q}}} - \Phi_{\underline{\mathbf{p}}q_4} \frac{\underline{\mathbf{q}}^T(t_0)}{q_4(t_0)} + \Phi_{\underline{\mathbf{p}}p_4} \frac{p_4(t_0)\underline{\mathbf{q}}(t_0)}{q_4^2(t_0)} - \Phi_{\underline{\mathbf{p}}p_4} \frac{\underline{\mathbf{p}}^T(t_0)}{q_4(t_0)} & \Phi_{\underline{\mathbf{p}}\underline{\mathbf{p}}} - \Phi_{\underline{\mathbf{p}}p_4} \frac{\underline{\mathbf{q}}^T(t_0)}{q_4(t_0)} \end{bmatrix}$$

$$\begin{Bmatrix} \delta \underline{\mathbf{q}}(t) \\ \delta \underline{\mathbf{p}}(t) \end{Bmatrix} = [\Phi_c(t, t_0)] \begin{Bmatrix} \delta \underline{\mathbf{q}}(t_0) \\ \delta \underline{\mathbf{p}}(t_0) \end{Bmatrix} \quad (133)$$

The 6×6 matrix $[\Phi_c(t, t_0)]$ in equation (133) represents the constrained Jacobian and the nonlinearity index is concerned with the worst case variation of $[\Phi_c(t, t_0)]$, not $[\Phi(t, t_0)]$.

Study of the Nonlinearity Index for Various Attitude Dynamics Coordinates

The nominal angular motions for simulation purposes is defined such that all the attitude parameterizations are far away from their singularities. Figures 14 and 15 show the time histories of nominal angular velocity and the nominal Euler angles. To consider the effect of various attitude parameters on the validity of linear error propagation, N worst-case initial attitude errors are chosen uniformly on sphere of radius

10 degrees, i.e. $\delta\phi_0^2 + \delta\theta_0^2 + \delta\psi_0^2 = \left(\frac{10\Pi}{180}\right)^2$. The Euler angle initial conditions are

mapped consistently through the appropriate coordinate transformation to generate corresponding initial condition variations for the alternative representations. Similarly, N worst-case initial errors for angular velocity coordinates are chosen near-uniformly on a sphere of radius $10 \mu\text{radians/sec}$.

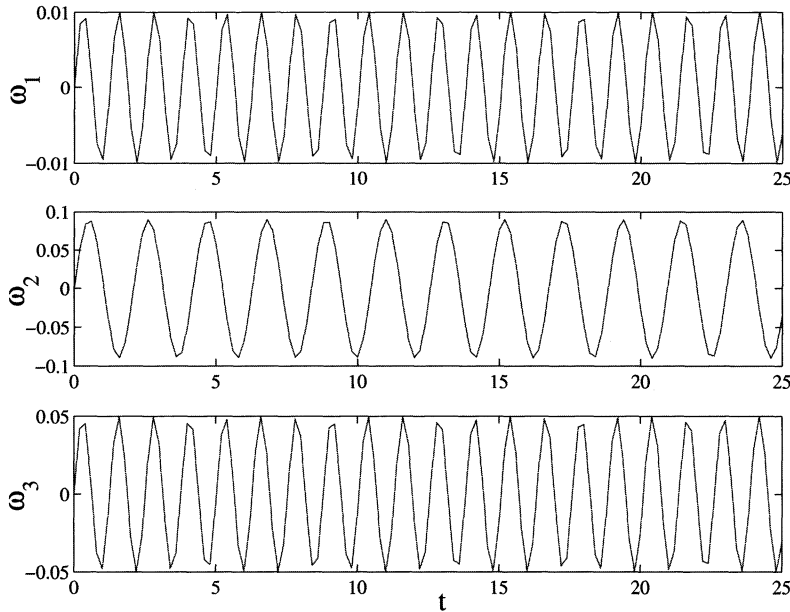


FIG. 14. Nominal Angular Velocity (in Rad/Sec).

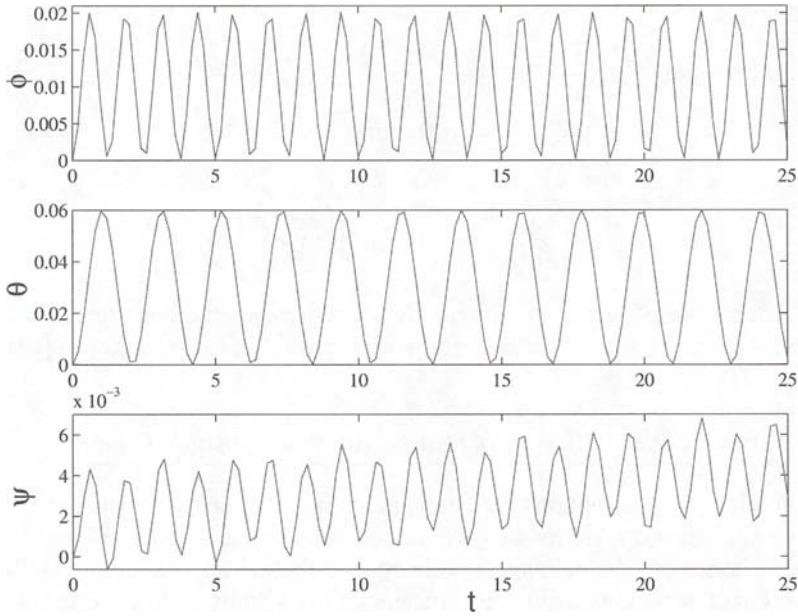


FIG. 15. Nominal Euler 3-2-1 Angles (in Radians).

Referring to Fig. 15, as a consequence of the nominal motion remaining near the ideal $(0, 0, 0)$ state to linearize 3-2-1 Euler angles, we do not expect large departures from the linear approximations. Figure 16 shows the variation of dynamic nonlinearity index for all attitude parameterizations with time. The computations of dynamic non-linearity index are based upon the 500 worst-case initial conditions. From Fig. 17, it is clear that Quaternions and Rodrigues parameters have linear approximation errors ($< 9\%$), whereas Modified Rodrigues parameters have linear

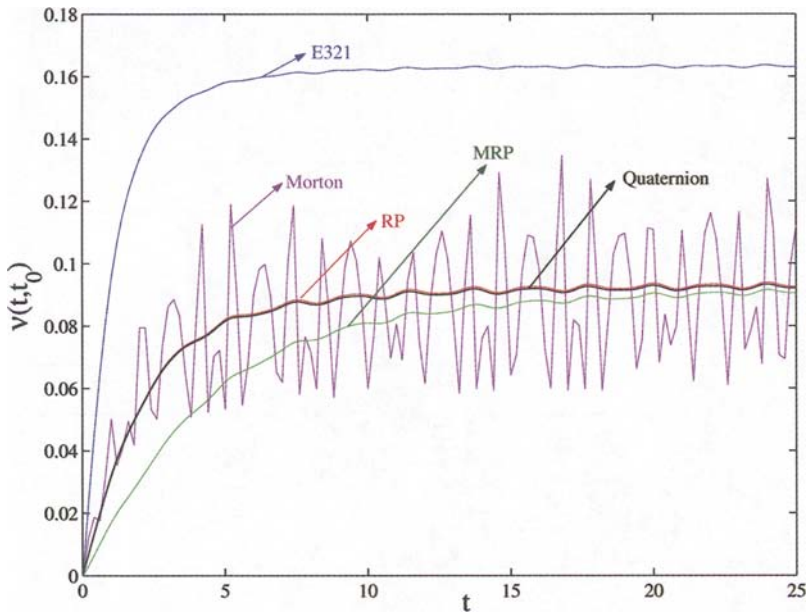


FIG. 16. Nonlinearity Index Variation with Time for Attitude Dynamics.

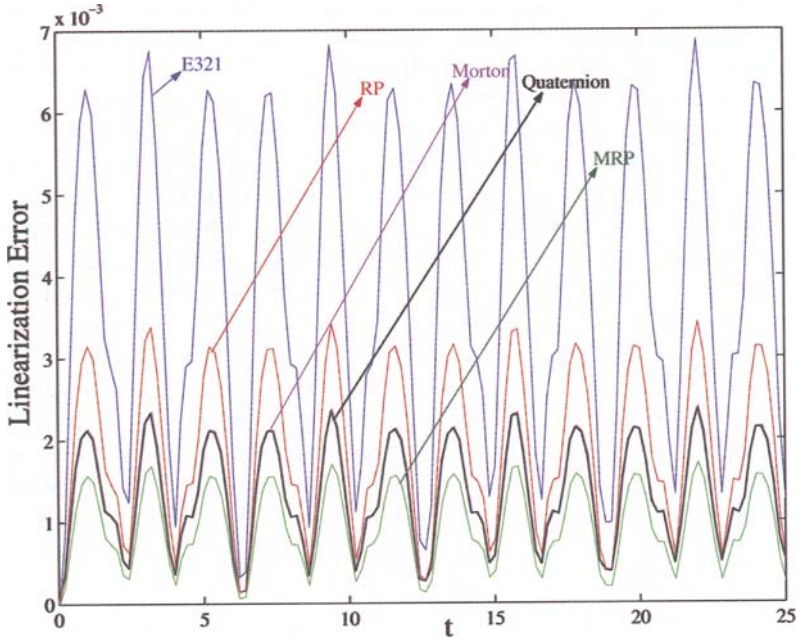


FIG. 17. Average Norm of Linearization Errors in Respective Attitude Parameter Space.

approximation errors ($< 8\%$) and Euler angles set have the worst linear approximation errors $> 15\%$. The canonical quaternion formulation of Morton have linear approximation errors ranging from $5\%–12\%$, for the selected nominal motion and variations in the neighborhood thereof. Also evident in Fig. 17, the actual linearization errors are in full agreement with the small $\nu(t, t_0)$ behavior. In Fig. 18,

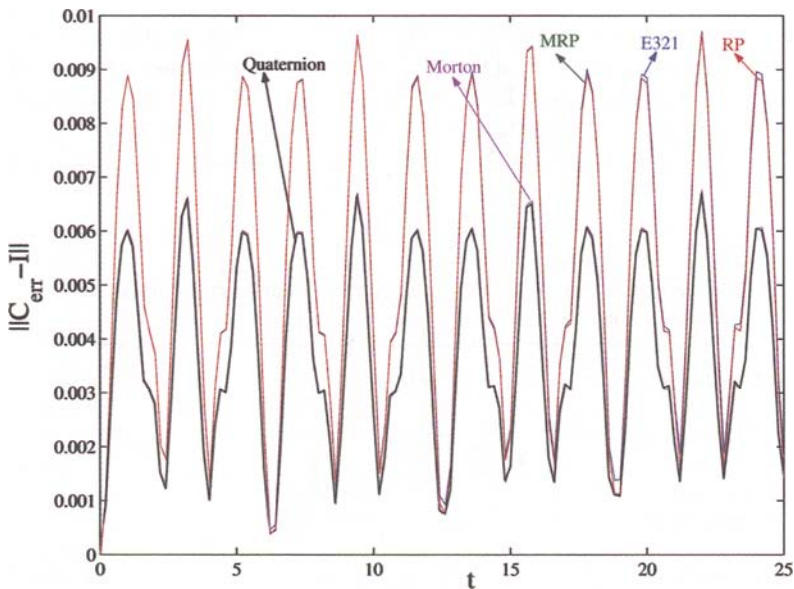


FIG. 18. Average Norm of the Linear Attitude Prediction Error for Various Representations.

the linear prediction errors of all coordinates are mapped nonlinearly for comparison into the error attitude matrix and compared to the attitude matrix corresponding to zero prediction error i.e. identity matrix. In Table 2, the values of average approximation errors for attitude parameters and angular velocities are given. As expected the angular velocity approximation errors are same for all the parameterizations except Morton's canonical formulation. It is evident from Fig. 18 and Table 2 that the attitude prediction errors for linearizing the Euler 3-2-1, RP, and MRP kinematics are of same order of magnitude but are more than that of corresponding to Quaternion or Morton's formulation. For this relatively smaller set of worst case variations, it is evident that all linearization errors are small. However, the nonlinearity index still shows advantages for the quaternion and Morton formulation. It is evident by inspection of the differential equations [equations (46), (47)] that Morton's formulation and quaternion formulation are regular, whereas all three parameter formulations contain singularities; therefore the canonical quaternion formulation can be readily used without "artwork" for tumbling motions, whereas all three parameter formulations must contain logic to work around the singularities. No claims of global optimality are made, for obvious reasons.

Orbital Dynamics

We consider here four formulations of the perturbed two body problem, namely:

- Classical Cartesian coordinates
- Burdet Regularization
- KS Regularization
- Quaternion Regularization

Analogous to the above developments for rotational dynamics, we consider a specific reference orbit and study the nonlinearity index $\nu(t, t_0)$ behavior for a family of worst-case variations centered on a reference orbit. From these studies it is possible to derive a specific judgement of the near-linearity in the large neighborhood of the reference orbit. While the conclusions can not be considered, a general methodology is illustrated. We know in advance that the nature of the perturbations and the orbital eccentricity are dominant considerations that will affect the generalization of interpretations from the results. In this case, we have studied motions with significant variations about an intermediate eccentricity orbit, and have restricted our attention

TABLE 2. Linearization Errors for Different Attitude Parameters for Attitude Dynamics Case

Attitude Representation	Attitude Error $\ C_{true}C_{app}^T - I\ $	Angular Velocity Error $\ \omega_t - \omega_{app}\ $
Euler Angles	5.3458e-03	8.2155e-03
Rodrigues Parameters	5.2853e-03	8.2155e-03
Modified Rodrigues Parameters	5.2819e-03	8.2155e-03
Quaternion	3.7177e-03	8.2155e-03
Morton's Formulation	3.7314e-03	8.2227e-03

to illustrate the methodology with nonlinear variations about one significant example orbit. Furthermore, we have not included any non-Keplerian perturbations in these developments.

In the interest of compactness, we do not include the Jacobians needed to compute the state transition matrix for each coordinate choice (analogous to those for the various attitude coordinates presented previously), rather we include these details in the report [34].

Summary of Orbital Mechanics for Unperturbed Motion

Rectangular Coordinates:

State Variables:

$$\mathbf{x} = \{x \ y \ z \ \dot{x} \ \dot{y} \ \dot{z}\}^T \quad (134)$$

State Equations of Motion:

$$\begin{aligned} \dot{\mathbf{x}} &= f(\mathbf{x}) \\ f(\mathbf{x}) &= \left\{ x_4 \ x_5 \ x_6 \ -\frac{\mu x_1}{r^3} + a_{dx} \ -\frac{\mu x_2}{r^3} + a_{dy} \ -\frac{\mu x_3}{r^3} + a_{dz} \right\}^T \end{aligned} \quad (135)$$

Departure Motion Differential Equations:

$$\delta \dot{\mathbf{x}} = A \delta \mathbf{x}, A = \begin{bmatrix} O_{3 \times 3} & I_{3 \times 3} \\ G & O_{3 \times 3} \end{bmatrix} + \frac{\partial}{\partial \mathbf{x}} \left\{ O_{3 \times 1} \right\} \begin{bmatrix} \mathbf{a}_d \end{bmatrix} \quad (136)$$

where

$$G = \frac{\mu}{r^5} \begin{bmatrix} 3x_1 - r^2 & 3x_1x_2 & 3x_1x_3 \\ 3x_1x_2 & 3x_2 - r^2 & 3x_2x_3 \\ 3x_1x_3 & 3x_2x_3 & 3x_3 - r^2 \end{bmatrix}$$

State Transition Matrix Differential Equation:

$$\dot{\Phi} = A\Phi, \Phi(t, t_0) = I \quad (137)$$

If $\mathbf{a}_d$ is zero, of course, there is an analytical solution for the state transition matrix [2].

KS Regularization:

State Variables:

$$\mathbf{x} = \{u_1 \ u_2 \ u_3 \ u_4 \ u'_1 \ u'_2 \ u'_3 \ u'_4 \ \alpha\}^T \quad (138)$$

State Equations of Motion:

$$\mathbf{x}' = \{x_5 \ x_6 \ x_7 \ x_8 \ -\alpha x_1 \ -\alpha x_2 \ -\alpha x_3 \ -\alpha x_4 \ 0\}^T \quad (139)$$

$$\frac{ds}{dt} = \frac{1}{\mathbf{u}^T \mathbf{u}} \quad (140)$$

State Transition Matrix Differential Equation:

$$\dot{\Phi} = A\Phi, A(s) = \begin{bmatrix} O_{4 \times 4} & I_{4 \times 4} & O_{1 \times 4} \\ -\alpha I_{4 \times 4} & O_{4 \times 4} & O_{1 \times 4} \\ 0 & 0 & 0 \end{bmatrix} \quad (141)$$

Equation (141) suggests that arbitrary initial condition variation of the KS coordinates are mapped into instantaneous KS variations as

$$\delta \mathbf{x}(s) = \Phi(s, s_0) \delta \mathbf{x}(s_0) \quad (142)$$

However, the independent variable, modified eccentric anomaly, s , itself is dependent upon time according to equation (140). Therefore, if we want to find the solution of differential equation (139) corresponding to a given time vector then we should assume s as a function of initial conditions which can result in a different differential equation for the state transition matrix Φ .

Let us consider the differential equation governing the state vector of equation (139):

$$\mathbf{x}' = \mathbf{f}(s, \mathbf{x}(s)) \quad (143)$$

Integration of this differential equation yields

$$\mathbf{x}(s) = \mathbf{x}(0) + \int_0^s \mathbf{f}(s, \mathbf{x}(s)) ds \quad (144)$$

Now, differentiating equation (144) with respect to the initial state vector $\mathbf{x}(0)$ yields the expression for state transition matrix $\Phi(s, s_0)$ as

$$\Phi(s, s_0) = \frac{\partial \mathbf{x}(s)}{\partial \mathbf{x}(0)} = I_{n \times n} + \mathbf{f}(s, \mathbf{x}(s)) \frac{\partial s}{\partial \mathbf{x}(0)} + \int_0^s \frac{\partial \mathbf{f}(\cdot)}{\partial \mathbf{x}(\cdot)} \frac{\partial \mathbf{x}(s)}{\partial \mathbf{x}(0)} ds \quad (145)$$

Let us introduce the notation

$$\begin{aligned} A(s) &= \frac{\partial \mathbf{f}(\cdot)}{\partial \mathbf{x}(s)} \\ \Psi(s) &= \frac{\partial s}{\partial \mathbf{x}(0)} \end{aligned} \quad (146)$$

Using the expressions given by equation (146), we can rewrite equation (145) as

$$\Phi(s, s_0) = I_{n \times n} + \mathbf{f}(s, \mathbf{x}(s)) \Psi(s) + \int_0^s A(s) \frac{\partial \mathbf{x}(s)}{\partial \mathbf{x}(0)} ds \quad (147)$$

Differentiating expression (147) with respect to s yields the expression for state transition matrix as

$$\Phi'(s, s_0) = \frac{\partial \Phi(s, s_0)}{\partial s} = \mathbf{f}(s, \mathbf{x}(s)) \frac{\partial \Psi(s)}{\partial s} + \frac{\partial \mathbf{f}(\cdot)}{\partial s} \Psi(s) + A(s) \Phi(s, s_0), \quad (148)$$

$$\Phi(s_0, s_0) = I_{n \times n}, \Psi(s_0) = O_{1 \times n}$$

Further, to calculate Ψ , let us consider the expression

$$t = t_0 + \int_0^s \mathbf{x}_{14}(s)^T \mathbf{x}_{14}(s) ds \quad (149)$$

where $\mathbf{x}_{14}$ is a short-hand notation for $\{x_1 \ x_2 \ x_3 \ x_4\}^T$; in general, $\mathbf{x}_{ij}$ represents $\{x_i \ x_{i+1} \ \dots \ x_j\}^T$. Now, differentiating equation (149) w.r.t. the initial condition vector $\mathbf{x}_0$ yields

$$\frac{\partial s}{\partial \mathbf{x}_0} \mathbf{x}_{14}(s)^T \mathbf{x}_{14}(s) + 2 \int_0^s \mathbf{x}_{14}^T(s) \frac{\partial \mathbf{x}_{14}(s)}{\partial \mathbf{x}_0} ds = 0 \quad (150)$$

Solving the above equation for $\frac{\partial s}{\partial \mathbf{x}_0}$ yields

$$\Psi(s) = \frac{\partial s}{\partial \mathbf{x}_0} = - \frac{2}{\mathbf{x}_{14}(s)^T \mathbf{x}_{14}(s)} \int_0^s \mathbf{x}_{14}^T(s) \frac{\partial \mathbf{x}_{14}(s)}{\partial \mathbf{x}_0} ds \quad (151)$$

Differentiating the expression for Ψ w.r.t. s yields the differential equation for Ψ as

$$\begin{aligned} \Psi'(s) &= \frac{4}{(\mathbf{x}_{14}(s)^T \mathbf{x}_{14}(s))^2} \mathbf{x}_{14}^T(s) \frac{\partial \mathbf{x}_{14}(s)}{\partial s} \int_0^s \mathbf{x}_{14}^T(s) \frac{\partial \mathbf{x}_{14}(s)}{\partial \mathbf{x}_0} ds \\ &\quad - \frac{2}{\mathbf{x}_{14}(s)^T \mathbf{x}_{14}(s)} \mathbf{x}_{14}^T(s) \frac{\partial \mathbf{x}_{14}(s)}{\partial \mathbf{x}_0} \\ &= - \frac{2}{\mathbf{x}_{14}^T \mathbf{x}_{14}} \mathbf{x}_{14}^T \left(\frac{\partial \mathbf{x}_{14}}{\partial s} \Psi + \frac{\partial \mathbf{x}_{14}}{\partial \mathbf{x}_0} \right) \end{aligned} \quad (152)$$

Now, using equations (148) and (152), we can calculate the unconstrained state transition matrix for KS-regularization.

Analogous to the developments for the redundant attitude dynamics coordinates, the above system is of ninth order, and we must constrain admissible departure motion to be consistent with the fact that the minimal order description is of sixth order. Therefore, initial condition variations must satisfy the following constraints:

1. Definition of α :

$$x_9 \equiv \alpha = \frac{1}{\mathbf{x}_{14}^T \mathbf{x}_{14}} \left(\frac{\mu}{2} - \mathbf{x}_{58}^T \mathbf{x}_{58} \right) \quad (153)$$

2. Initial condition constraint (assuming $x > 0$):

$$x_4 = 0 \quad (154)$$

3. Bilinear Relationship:

$$x_4 x_5 - x_3 x_6 + x_2 x_7 - x_1 x_8 = 0 \quad (155)$$

To find the constrained state transition matrix, we need to take the variations of constraint equations (153–155). The first variation of equations (153–155) yields

$$\delta x_9 = -\frac{2}{\mathbf{x}_{14}^T \mathbf{x}_{14}} \left[\frac{\mathbf{x}_{14}^T \delta \mathbf{x}_{14}}{\mathbf{x}_{14}^T \mathbf{x}_{14}} \left\{ \frac{\mu}{2} - \mathbf{x}_{58}^T \mathbf{x}_{58} \right\} + \mathbf{x}_{58}^T \delta \mathbf{x}_{58} \right] \quad (156)$$

$$\delta x_4 = 0 \quad (157)$$

$$\delta x_8 = \frac{1}{x_1} \{-x_3 \delta x_6 - x_6 \delta x_3 + x_2 \delta x_7 + x_7 \delta x_2 - x_8 \delta x_1\} \quad (158)$$

Rewriting equations (156–158) in terms of δx_{13} and δx_{57} gives

$$\delta x_4 = 0 \quad (159)$$

$$\begin{aligned} \delta x_8 &= \frac{1}{x_1} \{[-x_8 \quad x_7 \quad -x_6]^T \delta \mathbf{x}_{13} + \{0 \quad -x_3 \quad x_2\}^T \delta \mathbf{x}_{57}\} \\ &= \mathbf{a}^T \delta \mathbf{x}_{13} + \mathbf{b}^T \delta \mathbf{x}_{57} \end{aligned} \quad (160)$$

$$\begin{aligned} \delta x_9 &= -\frac{2}{\mathbf{x}_{13}^T \mathbf{x}_{13}} \left[\left(\frac{1}{\mathbf{x}_{13}^T \mathbf{x}_{13}} \left\{ \frac{\mu}{2} - \mathbf{x}_{58}^T \mathbf{x}_{58} \right\} \mathbf{x}_{13}^T + \mathbf{a}^T x_8 \right) \delta \mathbf{x}_{13} + (\mathbf{x}_{57}^T + x_8 \mathbf{b}^T) \delta \mathbf{x}_{57} \right] \\ &= \mathbf{c}^T \delta \mathbf{x}_{13} + \mathbf{d}^T \delta \mathbf{x}_{57} \end{aligned} \quad (161)$$

It is useful to partition Φ as

$$\Phi(s, s_0) = \begin{bmatrix} \Phi_{\mathbf{x}_{13}\mathbf{x}_{13}} & \Phi_{\mathbf{x}_{13}x_4} & \Phi_{\mathbf{x}_{13}\mathbf{x}_{57}} & \Phi_{\mathbf{x}_{13}x_8} & \Phi_{\mathbf{x}_{13}x_9} \\ \Phi_{x_4\mathbf{x}_{13}} & \Phi_{x_4x_4} & \Phi_{x_4\mathbf{x}_{57}} & \Phi_{x_4x_8} & \Phi_{x_4x_9} \\ \Phi_{\mathbf{x}_{57}\mathbf{x}_{13}} & \Phi_{\mathbf{x}_{57}x_4} & \Phi_{\mathbf{x}_{57}\mathbf{x}_{57}} & \Phi_{\mathbf{x}_{57}x_8} & \Phi_{\mathbf{x}_{57}x_9} \\ \Phi_{x_8\mathbf{x}_{13}} & \Phi_{x_8x_4} & \Phi_{x_8\mathbf{x}_{57}} & \Phi_{x_8x_8} & \Phi_{x_8x_9} \\ \Phi_{x_9\mathbf{x}_{13}} & \Phi_{x_9x_4} & \Phi_{x_9\mathbf{x}_{57}} & \Phi_{x_9x_8} & \Phi_{x_9x_9} \end{bmatrix} \quad (162)$$

where $\Phi_{\mathbf{x}_{ii}\mathbf{x}_{kl}}$ is a short-hand notation for $\Phi_{i \rightarrow j, k \rightarrow l}$. Now, from the definition of state transition matrix, we have

$$\delta \mathbf{x}(s) = \Phi(s, s_0) \delta \mathbf{x}(s_0) \quad (163)$$

where $\delta \mathbf{x} = [\delta \mathbf{x}_{13}^T \quad \delta x_4 \quad \delta \mathbf{x}_{57}^T \quad \delta x_8 \quad \delta x_9]^T$.

Using equations (159–161) and (162), we can write the constrained state transition matrix $\Phi_c(s, s_0)$ as

$$\Phi_c(s, s_0) = \begin{bmatrix} \Phi_{\mathbf{x}_{13}\mathbf{x}_{13}} + \Phi_{\mathbf{x}_{13}x_8} \mathbf{a}^T + \Phi_{\mathbf{x}_{13}x_9} \mathbf{c}^T & \Phi_{\mathbf{x}_{13}\mathbf{x}_{57}} + \Phi_{\mathbf{x}_{13}x_8} \mathbf{b}^T + \Phi_{\mathbf{x}_{13}x_9} \mathbf{d}^T \\ \Phi_{\mathbf{x}_{57}\mathbf{x}_{13}} + \Phi_{\mathbf{x}_{57}x_8} \mathbf{a}^T + \Phi_{\mathbf{x}_{57}x_9} \mathbf{c}^T & \Phi_{\mathbf{x}_{57}\mathbf{x}_{57}} + \Phi_{\mathbf{x}_{57}x_8} \mathbf{b}^T + \Phi_{\mathbf{x}_{57}x_9} \mathbf{d}^T \end{bmatrix} \quad (164)$$

Burdet's Regularization:

State Variables:

$$\mathbf{x} = \left\{ \frac{x}{r} \quad \frac{y}{r} \quad \frac{z}{r} \quad \left(\frac{x}{r} \right)' \quad \left(\frac{y}{r} \right)' \quad \left(\frac{z}{r} \right)' \quad \rho \quad \rho' \quad p \right\}^T \quad (165)$$

State Equations of Motion:

$$\mathbf{x}' = \left\{ x_4 \quad x_5 \quad x_6 \quad -x_1 \quad -x_2 \quad -x_3 \quad x_8 \quad \frac{1}{x_9} - x_7 \quad 0 \right\}^T \quad (166)$$

$$\frac{ds}{dt} = x_7^2 \sqrt{\mu x_9} \quad (167)$$

State Transition Matrix Differential Equation:

$$\dot{\Phi} = A\Phi \quad (168)$$

$$A(\phi) = \begin{bmatrix} O_{3 \times 3} & I_{3 \times 3} & O_{3 \times 3} \\ -I_{3 \times 3} & O_{3 \times 3} & O_{3 \times 3} \\ O_{3 \times 3} & O_{3 \times 3} & G \end{bmatrix}, \quad G = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & -\frac{1}{x_9^2} \\ 0 & 0 & 0 \end{bmatrix} \quad (169)$$

Equation (168) suggests that arbitrary initial condition variations of Burdet's coordinates are mapped into instantaneous Burdet variations as

$$\delta \mathbf{x}(\phi) = \Phi(\phi, \phi_0) \delta \mathbf{x}(\phi_0) \quad (170)$$

However, the independent variable true anomaly ϕ itself is dependent upon time according to equation (167). Therefore, if we want to find the solution of the differential equation (166) corresponding to a given time then we should assume ϕ as a function of initial conditions which can result in a different differential equation for state transition matrix Φ . Adopting the same procedure as we did in last section for KS variables, we can obtain the differential equation for the state transition matrix taking into account the time equation as well as

$$\Phi(\phi, \phi_0) = \frac{\partial \Phi(\phi, \phi_0)}{\partial \phi} = f(\phi, \mathbf{x}(\phi)) \frac{\partial \Psi(\phi)}{\partial \phi} + \frac{\partial f(\phi)}{\partial \phi} \Psi(\phi) + A(\phi) \Phi(\phi, \phi_0), \quad (171)$$

$$\Phi(\phi_0, \phi_0) = I_{n \times n}, \quad \Psi(\phi_0) = \mathbf{O}_{1 \times n}$$

Further, let us consider the time equation to compute Ψ given as

$$\frac{dt}{d\phi} = \frac{1}{x_7^2 \sqrt{\mu x_9}} \quad (172)$$

Integrating equation (172) w.r.t. ϕ yields

$$t = t_0 + \int_0^\phi \frac{1}{x_7^2 \sqrt{\mu x_9}} d\phi \quad (173)$$

Differentiation of equation (173) w.r.t. $\mathbf{x}_0$ yields

$$\frac{\partial \phi}{\partial \mathbf{x}_0} \frac{1}{x_7^2 \sqrt{\mu x_9}} + \int_0^\phi \left(-\frac{2}{x_7^3 \sqrt{\mu x_9}} \frac{\partial x_7}{\partial \mathbf{x}_0} - \frac{1}{2x_7^2 \sqrt{\mu x_9^3}} \frac{\partial x_9}{\partial \mathbf{x}_0} \right) d\phi = 0 \quad (174)$$

Solving equation (174) for $\frac{\partial \phi}{\partial \mathbf{x}_0}$ gives

$$\Psi = \frac{\partial \phi}{\partial \mathbf{x}_0} = x_7^2 \sqrt{\mu x_9} \int_0^\phi \frac{1}{x_7^2 \sqrt{\mu x_9}} \left(\frac{2}{x_7} \frac{\partial x_7}{\partial \mathbf{x}_0} + \frac{1}{2x_9} \frac{\partial x_9}{\partial \mathbf{x}_0} \right) d\phi \quad (175)$$

Now, differentiating equation (175) w.r.t. ϕ yields an expression for Ψ' as

$$\begin{aligned}\Psi' &= \left(2x_7 \sqrt{\mu x_9} \frac{\partial x_7}{\partial \phi} + \frac{1}{2} x_7^2 \sqrt{\frac{\mu}{x_9}} \frac{\partial x_9}{\partial \phi} \right) \int_0^\phi \frac{1}{x_7^2 \sqrt{\mu x_9}} \left(\frac{2}{x_7} \frac{\partial x_7}{\partial \phi} + \frac{1}{2x_9} \frac{\partial x_9}{\partial \phi} \right) d\phi \\ &\quad + \left(\frac{2}{x_7} \frac{\partial x_7}{\partial \phi} + \frac{1}{2x_9} \frac{\partial x_9}{\partial \phi} \right) \\ &= \left(\frac{2}{x_7} \frac{\partial x_7}{\partial \phi} + \frac{1}{2x_9} \frac{\partial x_9}{\partial \phi} \right) \Psi + \left(\frac{2}{x_7} \frac{\partial x_7}{\partial \phi} + \frac{1}{2x_9} \frac{\partial x_9}{\partial \phi} \right)\end{aligned}\quad (176)$$

Analogous to the developments for the redundant attitude dynamics coordinates, the above system is of ninth order and we must constrain admissible departure motion to be consistent with the truth that the minimal order description is of sixth order. Therefore, initial condition variations must satisfy the following constraints:

1. Norm Constraint:

$$x_1^2 + x_2^2 + x_3^2 = 1 \quad (177)$$

2. Definition of p :

$$\begin{aligned}\mu p &= r^2 \{ \mathbf{r}^T \dot{\mathbf{r}} - \dot{r}^2 \} \\ &= \frac{1}{\rho^2} \left\{ \left(-\frac{\dot{\rho} \mathbf{x}_{13}}{\rho^2} + \frac{\dot{\mathbf{x}}_{13}}{\rho} \right)^T \left(-\frac{\dot{\rho} \mathbf{x}_{13}}{\rho^2} + \frac{\dot{\mathbf{x}}_{13}}{\rho} \right) - \frac{1}{\rho^4} \dot{\rho}^2 \right\}\end{aligned}\quad (178)$$

Simplifying the above expression using the fact that $\mathbf{x}_{13}^T \mathbf{x}_{13} = 1$, we get

$$\mu p = \frac{\dot{\mathbf{x}}^T \dot{\mathbf{x}}}{\rho^4}$$

Converting the $()$ operator to the $()'$ operator yields

$$x_4^2 + x_5^2 + x_6^2 = 1 \quad (179)$$

3. Orthogonality Constraint:

$$x_1 x_4 + x_2 x_5 + x_3 x_6 = 0 \quad (180)$$

To find the constrained state transition matrix, we need to take the variations of constraint equations (177–180). The first variation of equations (177–180) yields

$$\delta x_1 = -\frac{1}{x_1} (x_2 \delta x_2 + x_3 \delta x_3) \quad (181)$$

$$\delta x_5 = -\frac{1}{x_5} (x_4 \delta x_4 + x_6 \delta x_6) \quad (182)$$

$$\delta x_4 = -\frac{1}{x_1} (x_4 \delta x_1 + x_2 \delta x_5 + x_5 \delta x_2 + x_3 \delta x_6 + x_6 \delta x_3) \quad (183)$$

Solving equations (182) and (183) for δx_4 and δx_2 yields

$$\delta x_4 = -\frac{1}{c} (a \delta x_2 + b \delta x_3 + d \delta x_6) \quad (184)$$

$$\delta x_5 = -\frac{1}{x_5} \left(-\frac{ax_4}{c} \delta x_2 - \frac{bx_4}{c} \delta x_3 + \left(x_6 - \frac{\delta x_4}{c} \right) \delta x_6 \right) \quad (185)$$

where

$$a = x_5 - \frac{x_4 x_2}{x_1} \quad (186)$$

$$b = x_6 - \frac{x_4 x_3}{x_1} \quad (187)$$

$$c = x_1 - \frac{x_4 x_2}{x_5} \quad (188)$$

$$d = x_3 - \frac{x_6 x_2}{x_5} \quad (189)$$

It is useful to partition Φ as

$$\Phi(\phi, \phi_0) = \begin{bmatrix} \Phi_{11} & \Phi_{12} & \Phi_{13} & \Phi_{14} & \Phi_{15} & \Phi_{16} & \Phi_{179} \\ \Phi_{21} & \Phi_{22} & \Phi_{23} & \Phi_{24} & \Phi_{25} & \Phi_{26} & \Phi_{279} \\ \Phi_{31} & \Phi_{32} & \Phi_{33} & \Phi_{34} & \Phi_{35} & \Phi_{36} & \Phi_{379} \\ \Phi_{41} & \Phi_{42} & \Phi_{43} & \Phi_{44} & \Phi_{45} & \Phi_{46} & \Phi_{479} \\ \Phi_{51} & \Phi_{52} & \Phi_{53} & \Phi_{54} & \Phi_{55} & \Phi_{56} & \Phi_{579} \\ \Phi_{61} & \Phi_{62} & \Phi_{63} & \Phi_{64} & \Phi_{65} & \Phi_{66} & \Phi_{679} \\ \Phi_{791} & \Phi_{792} & \Phi_{793} & \Phi_{794} & \Phi_{795} & \Phi_{796} & \Phi_{7979} \end{bmatrix} \quad (190)$$

As the constrained variations of initial conditions must satisfy equations (181), (184) and (185), therefore adopting the procedure listed in previous section we obtain the expression for the constrained state transition matrix Φ_c as

$$\Phi_{c_{k1}} = \Phi_{i2} - \Phi_{i1} \frac{x_2}{x_1} - \frac{a}{c} \Phi_{i4} + \frac{x_4 a}{x_5 c} \Phi_{i5} \quad (191)$$

$$\Phi_{c_{k2}} = \Phi_{i3} - \Phi_{i1} \frac{x_3}{x_1} - \frac{b}{c} \Phi_{i4} + \frac{x_4 b}{x_5 c} \Phi_{i5} \quad (192)$$

$$\Phi_{c_{k3}} = \Phi_{i6} - \frac{d}{c} \Phi_{i4} + \left(\frac{x_4 d}{x_5 c} - \frac{x_6}{x_5} \right) \Phi_{i5} \quad (193)$$

$$\Phi_{c_{k46}} = \Phi_{i79}, \{(k, i)\} \equiv \{(1, 2); (2, 3); (3, 6); (4, 7); (5, 8); (6, 9)\} \quad (194)$$

Quaternion Regularization:

State Variables:

$$\mathbf{x} = \{q_1 \quad q_2 \quad q_3 \quad q_4 \quad \rho \quad \rho' \quad p\}^T \quad (195)$$

State Equations of Motion:

$$\mathbf{x}' = \mathbf{f}(\mathbf{x}) = \left\{ \frac{1}{2}\omega_3 x_2 \quad -\frac{1}{2}\omega_3 x_1 \quad \frac{1}{2}\omega_3 x_4 \quad -\frac{1}{2}\omega_3 x_3 \quad x_6 \quad -x_5 + \frac{1}{x_7} \quad 0 \right\}^T \quad (196)$$

$$\frac{d\phi}{dt} = x_5^2 \sqrt{\mu x_7} \quad (197)$$

State Transition Matrix Differential Equation:

$$\dot{\Phi} = A\Phi \quad (198)$$

$$A = \begin{bmatrix} 0 & \frac{\omega_3}{2} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\omega_3}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\omega_3}{2} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\omega_3}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -\frac{1}{x_7^2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad (199)$$

Equation (198) suggests that arbitrary initial condition variations are mapped into instantaneous variations as

$$\delta \mathbf{x}(\phi) = \Phi(\phi, \phi_0) \delta \mathbf{x}(\phi_0) \quad (200)$$

As the independent variable for quaternion regularization is same as for Burdet's regularization, by adopting the same procedure as we did in last section for Burdet's variables we can obtain the differential equation for the state transition matrix taking into account the time equation as well, giving

$$\Phi'(\phi, \phi_0) = \mathbf{f}'(\phi) \frac{\partial \Psi(\phi)}{\partial \phi} + \frac{\partial \mathbf{f}(\phi)}{\partial \phi} \Psi(\phi) + A(\phi) \Phi(\phi, \phi_0), \quad \Phi(\phi_0, \phi_0) = I_{n \times n} \quad (201)$$

$$\Psi' = \left(\frac{2}{x_5} \frac{\partial x_5}{\partial \phi} + \frac{1}{2x_7} \frac{\partial x_7}{\partial \phi} \right) \Psi + \left(\frac{2}{x_5} \frac{\partial x_5}{\partial \mathbf{x}_0} + \frac{1}{2x_7} \frac{\partial x_7}{\partial \mathbf{x}_0} \right), \quad \Psi(\phi_0) = 0 \quad (202)$$

It is evident that these equations (196–201) must be modified appropriately for each perturbation considered. In the report [35], we consider these details and the corresponding numerical results for the frequently dominant J_2 perturbation.

Analogous to the developments for the redundant quaternion kinematics, the above system is of seventh order and we must constrain admissible departure motion to be consistent with the unit norm constraint on the quaternion.

Introducing the first three and last three elements of state vector $\mathbf{x}$ as $\mathbf{x}_{13}$ and $\mathbf{x}_{57}$ respectively, we write

$$\mathbf{x} = \{\mathbf{x}_{13}^T \quad x_4 \quad \mathbf{x}_{57}^T\}^T \quad (203)$$

Now the unit norm constraint on quaternion can be written as

$$\mathbf{x}_{13}^T \mathbf{x}_{13} + x_4^2 = 1 \quad (204)$$

Taking the first variation of equation (204), then it follows that the admissible variations must lie in the tangent plane defined by

$$\mathbf{x}_{13}^T \delta \mathbf{x}_{13} + x_4 \delta x_4 = 0 \quad (205)$$

It is useful to partition the state transition matrix Φ as

$$\Phi = \begin{bmatrix} \Phi_{\mathbf{x}_{13}\mathbf{x}_{13}} & \Phi_{\mathbf{x}_{13}x_4} & \Phi_{\mathbf{x}_{13}\mathbf{x}_{57}} \\ \Phi_{x_4\mathbf{x}_{13}} & \Phi_{x_4x_4} & \Phi_{x_4\mathbf{x}_{57}} \\ \Phi_{\mathbf{x}_{57}\mathbf{x}_{13}} & \Phi_{\mathbf{x}_{57}x_4} & \Phi_{\mathbf{x}_{57}\mathbf{x}_{57}} \end{bmatrix} \quad (206)$$

As the constrained variations of initial conditions must satisfy equation (205), therefore adopting the procedure given in the previous section we obtain the expression for constrained state transition matrix Φ_c as

$$\Phi_c = \begin{bmatrix} \Phi_{\mathbf{x}_{13}\mathbf{x}_{13}} - \Phi_{\mathbf{x}_{13}x_4} \frac{\mathbf{x}_{13}^T}{x_4} & \Phi_{\mathbf{x}_{13}\mathbf{x}_{57}} \\ \Phi_{\mathbf{x}_{57}\mathbf{x}_{13}} - \Phi_{\mathbf{x}_{57}x_4} \frac{\mathbf{x}_{13}^T}{x_4} & \Phi_{\mathbf{x}_{57}\mathbf{x}_{57}} \end{bmatrix} \quad (207)$$

Study of the Nonlinearity Index for Various Orbit Dynamics Coordinates

In this section, we present the simulation results for studied motions with significant variations about an intermediate eccentricity orbit and a high eccentricity orbit. The nominal orbits considered for this study are shown in Figure 19. With respect to classical elements both orbits differ only in terms of eccentricity. The classical elements of these two nominal orbits are shown in Table 3.

The worst-case initial condition variations considered to evaluate nonlinearity for both the nominal orbits were specified in rectangular coordinates and are given in Table 4.

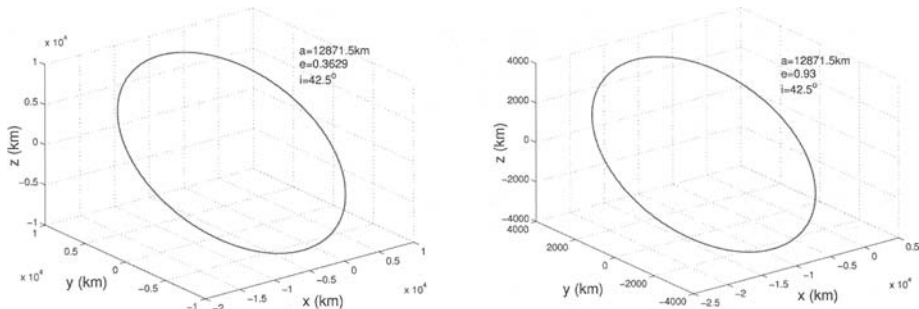


FIG. 19. Nominal Orbits.

TABLE 3. Classical Elements For Nominal Orbits

Orbit	a	e	i
Intermediate Eccentricity Orbit	12871.5 km	0.3629	42.5°
High Eccentricity Orbit	12871.5 km	0.93	42.5°

These variations give rise to a family of orbits with varying elements, however it was ensured that initial condition variations for the high eccentricity nominal orbit did not lead to parabolic or hyperbolic orbits.

To construct the worst-case surface (and the corresponding “uniformly spaced” worst-case initial conditions), the position and velocity variations are normalized by the respective worst-case values and the varied points are constrained to lie on the six-dimensional unit sphere. The uniformly spaced points on this sphere were used to generate initial conditions for the set of N rectangular coordinate simulations. Each of these points were mapped through the appropriate coordinate transformation into the corresponding initial conditions for the KS, Burdet and quaternion regularized formulations to generate the basis for computing the same N physical motions in the four coordinate systems.

To simplify the graphical studies, all results computed in the various coordinate choices are compared versus time, even though this is not the independent variable in several of the transformed coordinate spaces. The time differential equation is simply added to the state vector and it can be included in the nonlinearity index computation as described in previous section. It is of interest to see the difference in the nonlinearity index when the time equation is either excluded or included, because the time equation is usually “more nonlinear” than the state variables in the transformed spaces. Note that while initial conditions are treated as independent for the purpose of evaluating nonlinearity, in fact the redundant initial condition variations are constrained by the fact the transformed systems are in reality of sixth order; thus the redundant coordinates must be consistent. The constrained variation of worst-case initial conditions is accommodated in this case by prescribing the worst-case surface in a nonredundant coordinate system (here the Cartesian coordinates were used) and transforming the initial conditions into the appropriate redundant coordinate set using transformation equations that are properly constrained. Likewise, the linearity of dependent state coordinates is an issue that must be addressed in the context of the fact that these coordinates are redundant. For example, in solving a two-point boundary value problem, the initial conditions must be admissible and the specification of possibly redundant target boundary conditions must take full account of the constraints. In particular cases [36, 37], we may use Lagrange multipliers or constraint elimination to account for the redundant degrees of freedom. These comments broadly apply to both the redundant attitude and orbit formulations.

TABLE 4. Initial Condition Variations For Nominal Orbits

Orbit	$ \delta \mathbf{x} $	$ \delta \dot{\mathbf{x}} $
Intermediate Eccentricity Orbit	10 km	0.01 km/sec
High Eccentricity Orbit	1 km	0.001 km/sec

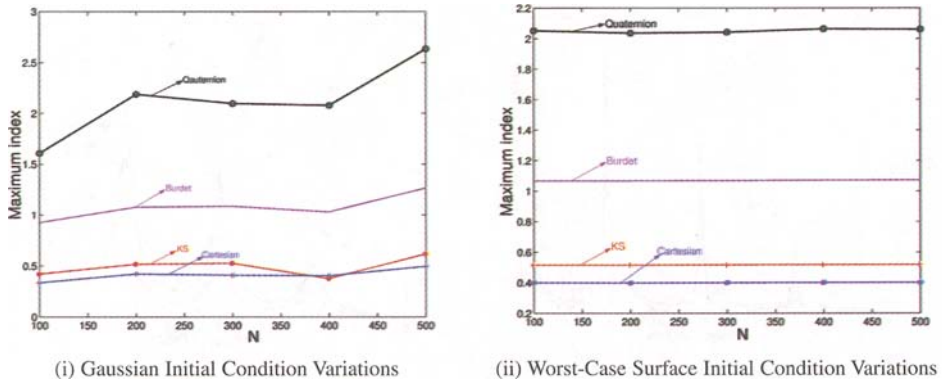


FIG. 20. Convergence of Nonlinearity Index for Various Orbit Coordinates (over Four Nominal Orbit Periods).

Figure 20 shows the convergence of nonlinearity index for several choices of orbit state variables with number of initial condition variation N . The nonlinearity index in Fig. 20 is computed for the intermediate eccentricity orbit taking into account the time equation as well for Burdet, KS and Quaternion regularization. It is clear in Fig. 20 that taking the worst-case initial conditions as uniformly spaced on the initial $3\text{-}\sigma$ surface [the six-dimensional ellipsoid is normalized to a sphere in principal axes by using the worst-case values (generally, three times the square root of the corresponding eigen-value of the covariance matrix)] results in dramatically faster convergence than does the usual Gaussian Monte Carlo method—in fact the uniform points lead to extremely efficient convergence in 100 to 200 trials, whereas, for the Monte Carlo method, convergence is not evident after 500 trials. The poor convergence in the Monte Carlo simulations is also due to the fact that sets of 100, 200, 300, 400 initial conditions are chosen independently from the set of 500 initial conditions (trials were not accumulated). We mention that the maximum value of ν found in the Monte Carlo analysis occasionally exceeds that in the worst-case initial conditions, because all points in the latter case are constrained to lie on the $3\text{-}\sigma$ surface, whereas, occasionally greater than $3\text{-}\sigma$ outliers arise in the Monte Carlo analysis.

Figure 21 shows the behavior of the nonlinearity index for several choices of orbit state variables by including and excluding the time equation for intermediate eccentricity orbit. It is evident in Fig. 21 that the KS formulation is most linear, whereas the Rectangular, Burdet, and Quaternion formulations are competitive when we exclude the time equation from our analysis. However, if we include the time equation in our analysis then KS and Rectangular coordinates are the winners but surprisingly Burdet and Quaternion are the worst because of their highly nonlinear time equation. As might be expected, all formulations behave most nonlinearly with respect to initial condition variations near successive perigees, and the long-term secular departure from linearity of the KS, Burdet, and Quaternion formulations is accelerated due to the implicit dependence on the parameters α and p on initial conditions. Surprisingly, the Burdet and Quaternion regularizations give almost identical nonlinearity indices. This is because the differential equation for $\rho = 1/r$ (common to both formulations and depends upon $1/p$, which depends nonlinearly on initial conditions) dominates the nonlinearity in both cases. Furthermore, even though the transformed differential equations are “rigorously

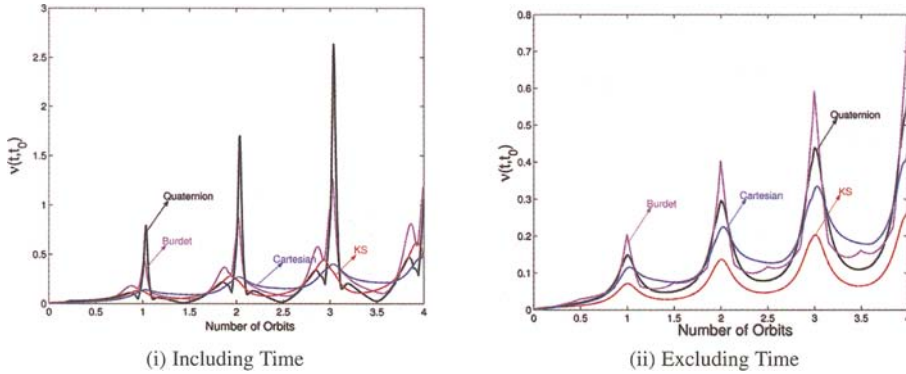


FIG. 21. Time Variation of Nonlinearity Index for Various Orbit Coordinates (over Four Nominal Orbit Periods).

linear” as a function of ϕ and p both implicitly depend nonlinearly on the initial conditions and the surprise is that these “linear” formulations are quite nonlinear, with respect to physical initial condition variations! They are even more nonlinear than the traditional rectangular coordinates. On the other hand, the KS regularization is less nonlinear than any of the formulations over a single orbit, in spite of the fact that the energy constant a (essentially the semimajor axis) also depends upon the initial conditions, and therefore the apparently linear solution is actually a nonlinear function of initial conditions. The KS advantages are severalfold; a crucial issue is that the generalized eccentric anomaly is attractive as compared to true anomaly, with regard to nonlinearity of the motion. Figure 22 shows the behavior of true and eccentric anomaly with respect to mean anomaly (essentially the normalized time). It is evident in Fig. 22 that as the eccentricity increases, the graph of true anomaly vs mean anomaly varies dramatically as a function of e (especially for $e > 0.6$), but less dramatic changes occur in eccentric anomaly.

Figure 23 shows the behavior of the nonlinear index for various orbit state variables by excluding the time equation for high eccentricity orbit. It is evident in Fig. 23 that KS, Burdet and Quaternion regularizations are more linear than the rectangular coordinates. This is due to the fact that for high eccentricity orbits, near

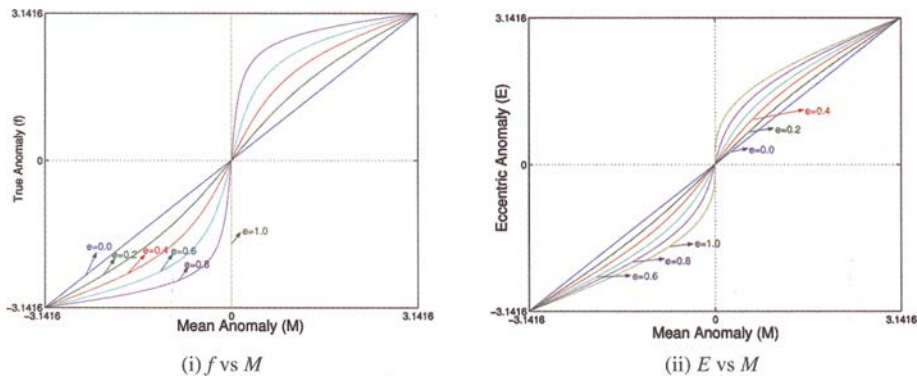


FIG. 22. Variation of True (f) and Eccentric Anomaly (E) with Mean Anomaly (M) for Different Eccentric Orbits.

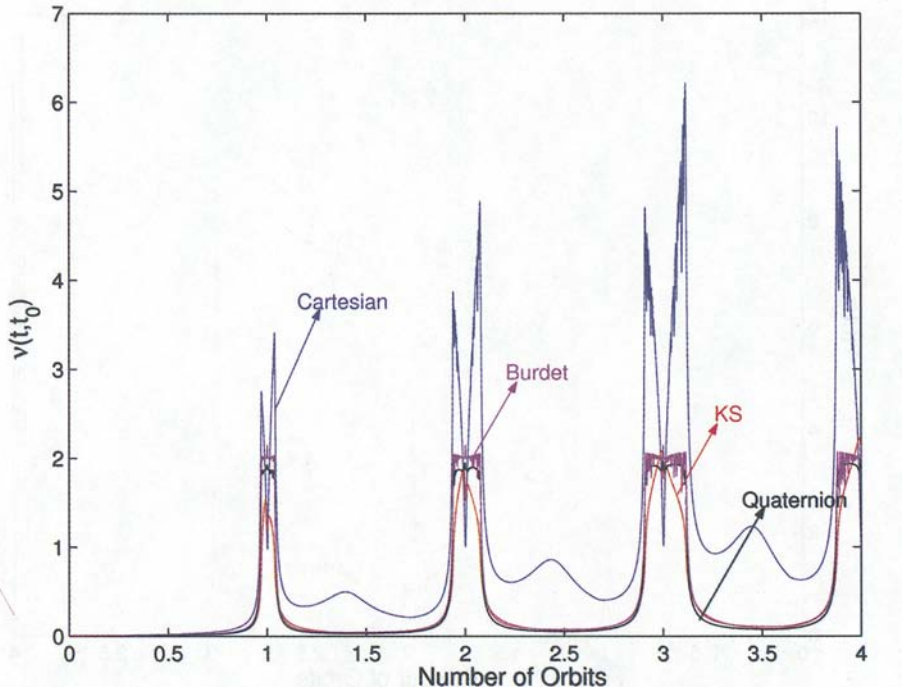


FIG. 23. Time Variation of Nonlinearity Index for Various Orbit Coordinates for High Eccentricity Orbit Excluding the Time Equation (over Four Nominal Orbit Periods).

the perigee the $1/r^2$ term (in the case of rectangular coordinates) “blows up” due to smaller value of r . For KS, Burdet, and Quaternion regularization the state variables are intelligently selected to avoid terms like $1/r^2$. However if we want to include the time equation in our analysis then it is clear from equations (152), (175), and (202) that we again encounter singular $1/r$ behavior. This is also evident from the nonlinearity index (including time equation) graph for KS and rectangular variables in Fig. 24. We did not plot the nonlinear indices for Burdet and Quaternion variables because they perform poorly as compared to KS and rectangular variables. Therefore, even though the differential equations for KS, Burdet and Quaternion regularizations appear “rigorously linear” as a function of ϕ and s , *they are even more nonlinear than the traditional rectangular coordinates* if we include the time equation as well.

We note that, depending upon the goals of a particular analysis, we may be able to do much of our work using either s or ϕ as the time-like variable, and only on occasion, have to address the nonlinear mapping into real time. For that reason it is important to retain the two sets of conclusions from Fig. 21.

These results are consistent with our early study [26] wherein eccentricity was varied over a wide range—the KS regularization is increasingly advantageous in comparison to other known approaches as eccentricity is increased toward unity without considering the time equation. In the earlier studies [26], both the KS and Burdet formulations were found to be much more efficient than rectangular coordinates, as regards numerical integration of J_2 -perturbed motion, especially for high eccentricity. Thus, apparently, the efficiency of numerical solution of the differential equations does not equate one-to-one with nonlinearity with respect to initial

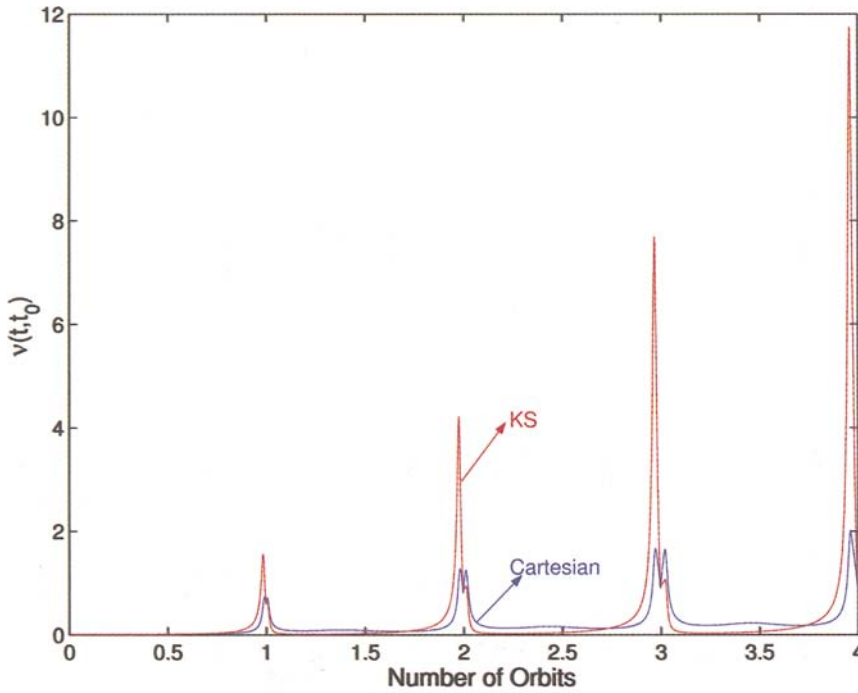


FIG. 24. Time Variation of Nonlinearity Index for Various Orbit Coordinates for High Eccentricity Orbit Including Time Equation (over Four Nominal Orbit Periods).

conditions. Of course all of these observations depend strongly on the perturbations included; the results of reference [26] include oblateness (J_2).

This paper illustrates the use of the nonlinearity index to evaluate competing coordinates, *vis a vis* nonlinearity, over a large family of motions belonging to a worst-case set centered on a particular reference motion. It is not an exhaustive study over all species of orbits as a function of energy, eccentricity, and for various perturbations in order to make broad conclusions, but rather an illustration of methodology. More comprehensive numerical studies can of course be conducted in future work.

Concluding Remarks

In this paper, we presented some tutorial and broadly applicable ideas on answering the title question “how nonlinear is it?” We began by defining measures of nonlinearity applicable to algebraic and dynamical systems. We also discussed several competing coordinate choices for the two central problems of astrodynamics, namely spacecraft attitude dynamics and orbital dynamics. One feature of this paper is the unified treatment of regularized attitude and orbital dynamics. In particular, we made some connections between rotational kinematics and orbital dynamics and discussed a quaternion regularization developed from the historical work of Burdet, Vitins et al.; we find this to be a very attractive regularization of orbital dynamics, although the nonlinearity analysis with respect to initial condition variation favors the KS formulation. The nonlinearity index analysis shows how worst-case initial condition variations, sampled near uniformly on a closed

n -dimensional surface, can be used to seek out the most nearly linear coordinates for a given motion and bounded variations about this motion. In particular, the convergence of the nonlinearity measure realized is very efficient and shown superior to Monte Carlo methods. The results provide qualitative support for the utility of the nonlinearity index and also provide insight to the relative nonlinearity of several competing formulations of attitude dynamics and orbital dynamics. The numerical studies presented are not considered comprehensive, but are believed to offer a useful illustration of the methodology. The approach and results of this paper have broad applicability to coordinate selection and linearization approximations in dynamical systems generally and in astrodynamics in particular.

Finally, we mention that, *taken as a set, the results of this paper not only address the question raised in the title, but also remind us of the eternal beauty of analytical dynamics.*

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Dedication

We dedicate this paper to our distinguished colleague: **Professor Vinod J. Modi.**

Vinod: We salute you. You lived a happy, productive, compassionate, and many-dimensional life with style and grace. While your life is now complete—your legacy will endure indefinitely.

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